Simple and general multi-interacting fluids in SymBoltz

David Figueruelo[1,2] and Herman Sletmoen[2]

[1] *Department of Theoretical Physics, University of the Basque Country UPV/EHU, 48080 Bilbao, Spain*
[2] *Institute of Theoretical Astrophysics, University of Oslo, N-0315 Oslo, Norway*

(Dated: November 5, 2026)

We implement a simple and general way of specifying arbitrary interactions between baryons, dark matter and dark energy in the Einstein–Boltzmann code SymBoltz. Due to the symbolic-numeric interface and approximation-free structure of SymBoltz, the model definition is remarkably compact and convenient to modify by specifying only the full equations. They are stiff, but stable and fast to integrate with an efficient implicit solver, particularly with the sparse and analytical ODE Jacobian generated from the symbolic equations. We first validate some models with only one interaction by comparing matter and CMB power spectra to those from modified versions of CLASS, finding 0.1%-1% agreement at most scales. Finally, we show how our framework makes it trivial to implement a model with interactions between all three species at the same time, which would require more work in CLASS. Our full model is specified in only around 300 lines of code and is easy to modify with other interactions or changes.

I. INTRODUCTION

SymBoltz[1] [1] is a recently introduced Einstein–Boltzmann solver written in the Julia programming language.

It is completely free of approximation schemes that complicate other codes such as CLASS [2, 3] and CAMB [4], and it has a symbolic-numeric interface that makes it convenient to specify and modify equations. These features are useful for implementing models with interactions between multiple species.

First, approximation-freeness means that only the full and stiff Einstein–Boltzmann equations have to be specified, and it is not necessary to derive and implement approximations to the equations in different regimes. In CLASS, for example, the tight-coupling approximation (TCA) [3] is mandatory to circumvent stiffness of the equations in the early universe. Interacting models generally introduce additional stiffness and modify the equations underlying the TCA, requiring new or modified approximation schemes. Other approximations such as the radiation streaming approximation (RSA), ultra-relativistic fluid approximation (UFA) and non-cold dark matter fluid approximation (NCDMFA) [3, 5] are in principle optional and can be disabled, but are often necessary in practice because the code relies on them for the speed of solving the perturbation equations. SymBoltz instead uses the efficient implicit Rodas5P ODE solver [6] to handle stiffness, and provides it with fast numerical code for the ODE and its sparse and analytical Jacobian generated by the symbolic-numeric interface. The bottom line is that we can directly implement the full modified equations, without spending any effort on re-deriving, re-implementing and re-validating approximation schemes that are valid in different regimes.

Second, the symbolic-numeric interface allows for a convenient, compact and general way of specifying our modified equations. Our full ΛCDM model with interactions is encoded in only around 300 lines of code in the code documentation[2] and is easy to modify. All variables, parameters and equations for the background and perturbations and for all components are specified in a single place using Unicode characters with one-to-one correspondence with the mathematical notation. The symbolic-numeric engine automatically separates the background and perturbation, splines the background when solving the perturbations, generates optimal numerical code for the ODE and its Jacobian and makes it trivial to obtain any variables from the solution. Moreover, SymBoltz has been updated to version 2 alongside this paper, which makes it possible to integrate the background equations using a mixture of forwards and backwards integration in time. This is particularly useful for interacting models, where the background continuity equations (3.21) generally loses its analytical solution, and it is useful to integrate them backwards from present values of the density parameters.

II. SYMBOLTZ VERSION 2: MIXED BACKWARD-FORWARD INTEGRATION

SymBoltz version 2 is released alongside this paper. In particular, it updates the symbolic-numeric framework to let users easily specify models where some background[3] equations should be integrated forwards in time, while others

[1] https://github.com/hersle/SymBoltz.jl/
[2] https://hersle.github.io/SymBoltz.jl/stable/interacting/
[3] By background we mean everything to 0th order in perturbation theory. This includes what is referred to as thermodynamics in CLASS.

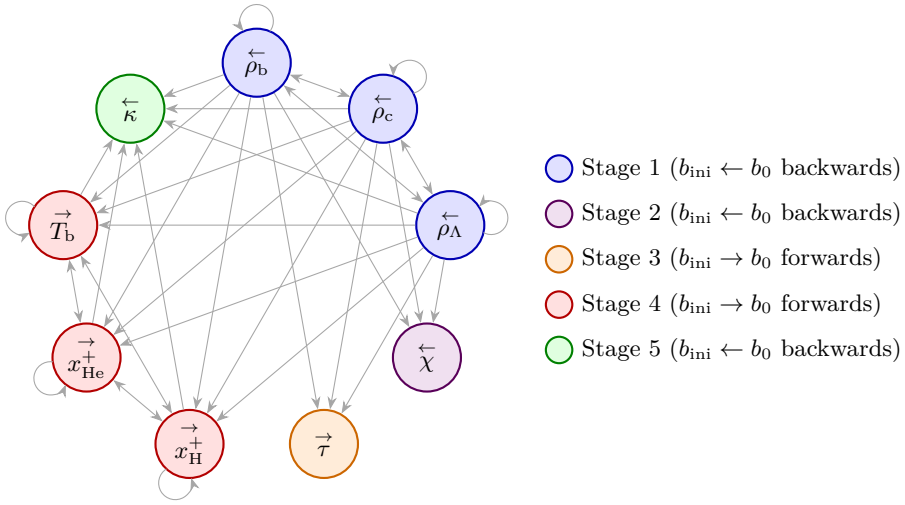


FIG. 1. SymBoltz version 2 automatically creates a dependency graph between all variables in the user-specified background equations. An arrow $x \to y$ means that $\dot{y}$ (and hence y) depends on x. SymBoltz identifies cycles in this graph (here with the same color), which must be integrated as one coupled ODE. Different cycles are disconnected or connected by one-way arrows, which constitute stages of the equations that can be integrated sequentially, each using splined input of earlier stages. Variables without outgoing arrows correspond to integrals. Each variable is marked with a preferred integration direction (forwards $\to$ or backwards $\leftarrow$), and all variables in the same coupled stage must have the same direction. Here the background system (2.1) admits 5 stages (going clockwise from 12 o'clock), and stages $1+2$ and $3+4$ can be merged because they share direction.

should be integrated backwards in. For example, the background ODEs (including thermodynamics and RECFAST recombination) we consider here have the following structure ($t = \log a$ is the independent time variable and $' = \mathrm{d}/\mathrm{d}t$ is its derivative):

Baryon density:	$\rho_\mathrm{b}{}' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda)$	integrated from	$\rho_\mathrm{b}(t_0)$	and backwards,	(2.1a)
Dark matter density:	$\rho_\mathrm{c}{}' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda)$	integrated from	$\rho_\mathrm{c}(t_0)$	and backwards,	(2.1b)
Dark energy density:	$\rho_\Lambda{}' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda)$	integrated from	$\rho_\Lambda(t_0)$	and backwards,	(2.1c)
Lookback time:	$\chi' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda)$	integrated from	$\chi(t_0)$	and backwards,	(2.1d)
Conformal time:	$\tau' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda)$	integrated from	$\tau(t_\mathrm{ini})$	and forwards,	(2.1e)
Hydrogen ionization:	$x^+_\mathrm{H}{}' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda, x^+_\mathrm{H}, x^+_\mathrm{He}, T_\mathrm{b})$	integrated from	$x^+_\mathrm{H}(t_\mathrm{ini})$	and forwards,	(2.1f)
Helium ionization:	$x^+_\mathrm{He}{}' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda, x^+_\mathrm{H}, x^+_\mathrm{He}, T_\mathrm{b})$	integrated from	$x^+_\mathrm{He}(t_\mathrm{ini})$	and forwards,	(2.1g)
Baryon temperature:	$T_\mathrm{b}{}' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda, x^+_\mathrm{H}, x^+_\mathrm{He}, T_\mathrm{b})$	integrated from	$T_\mathrm{b}(t_\mathrm{ini})$	and forwards,	(2.1h)
Optical depth:	$\kappa' = f(t, \rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda, x^+_\mathrm{H}, x^+_\mathrm{He}, T_\mathrm{b})$	integrated from	$\kappa(t_0)$	and backwards.	(2.1i)

The continuity equations for $i \in \{b, c, \Lambda\}$[4] are numerically integrated backwards because we parameterise them by their present density parameters Ω_{b0}, Ω_{c0} and $\Omega_{\Lambda 0}$. Conversely, thermodynamics equations are integrated forwards because their fully ionised initial conditions are in the early universe. This is a generalization of the equations for the standard ΛCDM model, where the non-interacting backwards continuity equations have analytical solutions $\rho_i(t) = \rho_i(t_0) a(t)^{-3(1+w_i)}$ with constant equations of state w_i. As illustrated in Fig. 1, the background system (2.1) admits five stages and the best way to integrate them is to join stages $1+2$ and $3+4$:

$$\text{integrate } \{\rho_\mathrm{b}, \rho_\mathrm{c}, \rho_\Lambda, \chi\} \text{ backwards} \quad \to \quad \text{integrate } \{\tau, x^+_\mathrm{He}, x^+_\mathrm{H}, T_\mathrm{b}\} \text{ forwards} \quad \to \quad \text{integrate } \{\kappa\} \text{ backwards.} \tag{2.2}$$

With the latest SymBoltz update, the user only needs to write down the full background (and perturbation) equations together, and the symbolic-numeric framework automatically inspects the symbolic equations to obtain the dependency graph between all variables and then splits them into appropriate stages. This can be understood as a generalisation of the `background.c` $\to$ `thermodynamics.c` $\to$ `...` structure in CLASS [2], such that the number of

[4] The label Λ denotes dark energy of any form or a cosmological constant. Although an abuse of notation, we use it to keep as close as possible to the internal notation of SymBoltz.

background stages is not hardcoded to 2. We emphasise that this machinery is not custom-tailored to this particular model, but made to be general with the ambition of supporting equations with arbitrary inter-dependencies and any mixture of forward-backward integration. For example, CLASS has special support a non-interacting continuity equation with $w = w(a)$ by first evaluating $\rho(a_{\rm ini}) = \rho(1)\exp\left(-3\int_0^{a_{\rm ini}} \mathrm{d}a\,\big(1+w(a)\big)/a\right)$, and then evolving the continuity equation forwards as an ODE. This falls naturally into the general backwards-forwards integration framework in SymBoltz by first doing backwards integration on the integral.

The alternative to integrating backwards is to only integrate forwards, but use the shooting method to find the initial conditions at $t_{\rm ini}$ that give the desired densities at t_0. This is also supported by SymBoltz. This amounts to solving the following root-finding problem for $\rho_{\rm b}(t_{\rm ini})$, $\rho_{\rm c}(t_{\rm ini})$ and $\rho_\Lambda(t_{\rm ini})$, where $\rho_{\rm b}(t_0)$, $\rho_{\rm c}(t_0)$ and $\rho_\Lambda(t_0)$ are computed by integrating an ODE forwards:

$$\mathbf{F}\big(\rho_{\rm b}(t_{\rm ini}), \rho_{\rm c}(t_{\rm ini}), \rho_\Lambda(t_{\rm ini})\big) = \left[\tfrac{8\pi}{3}\rho_{\rm b}(t_0) - \Omega_{b0},\ \tfrac{8\pi}{3}\rho_{\rm c}(t_0) - \Omega_{c0},\ \tfrac{8\pi}{3}\rho_\Lambda(t_0) - \Omega_{\Lambda 0}\right] = \mathbf{0}. \tag{2.3}$$

Here the differentiability of SymBoltz is used to compute the Jacobian $\mathbf{F}$ to help Newton methods converge to the right root. Whether backwards integration or shooting is better is model-dependent and comes down to where the initial conditions are known, the stability of the integration direction of the equations and the stability of the parameter space of the model. For this model we find backward and forward integration of the continuity equations to be equally stable, but with the shooting method we find it challenging to find initial guesses for $\rho_i(t_{\rm ini})$ that avoids running into physically unstable regions in parameter space (e.g. with negative energy densities) for general interactions. Our preferred approach is therefore backwards integration to avoid shooting.

The ambition with SymBoltz version 2 is that mixed backwards-forward integration in conjunction with the shooting method constitutes a framework for treating the boundary-value nature of the background equations for many cosmological models in a general way.

III. COVARIANTISATION OF COUPLINGS

A. Conventions and notation

We start from the most general linearly perturbed FLRW metric,

$$\mathrm{d}s^2 = a^2(\tau)\left[-(1+2A)\mathrm{d}\tau^2 - 2B_i\mathrm{d}\tau\mathrm{d}x^i + \left((1-D)\delta_{ij} + 2E_{ij}\right)\mathrm{d}x^i\mathrm{d}x^j\right]. \tag{3.1}$$

Here τ is conformal time, $\mathcal{H} \equiv a'/a$ is the conformal Hubble rate, and primes denote derivatives with respect to τ. Spatial indices are denoted by $i, j, \ldots$, while matter components are labelled by indices in parentheses, $(n), (m), \ldots$. The shift and spatial perturbation are decomposed as

$$\begin{aligned} B_i &= -\partial_i B^{(s)} + B_i^{(v)}, \qquad \text{with} \quad \partial_i B^{i(v)} = 0, \\ E_{ij} &= E_{ij}^{(s)} + E_{ij}^{(v)} + E_{ij}^{(t)}, \quad \text{with} \quad \begin{cases} E_{ij}^{(s)} = \left(\partial_i\partial_j - \frac{1}{3}\delta_{ij}\nabla^2\right)E, \\ E_{ij}^{(v)} = -\frac{1}{2}(\partial_i E_j + \partial_j E_i), \quad \partial_i E^i = 0, \\ \partial^i E_{ij}^{(t)} = 0, \quad \delta^{ij}E_{ij}^{(t)} = 0. \end{cases} \end{aligned} \tag{3.2}$$

For the matter sector we use, for a generic component (n),

$$T^0{}_{0(n)} = -(\bar\rho_{(\rm n)} + \delta\rho_{(\rm n)}), \tag{3.3}$$

$$T^0{}_{i(n)} = (\bar\rho_{(\rm n)} + \bar P_{(\rm n)})(v_{i(n)} - B_i), \tag{3.4}$$

$$T^i{}_{0(n)} = -(\bar\rho_{(\rm n)} + \bar P_{(\rm n)})v^i_{(\rm n)}, \tag{3.5}$$

$$T^i{}_{j(n)} = (\bar P_{(\rm n)} + \delta P_{(\rm n)})\delta^i{}_j + \bar P_{(\rm n)}\Pi^i{}_{j(n)}. \tag{3.6}$$

In what follows we consider perfect fluids and set $\Pi^i{}_{j(n)} = 0$ for convenience while this scheme can be extended to having anisotropic stresses. At linear order scalar, vector and tensor perturbations evolve independently. Here, we are going to restrict ourselves to scalar perturbations and work in the Newtonian gauge, for which we have $B_i = 0$ and $E_{ij} = 0$, so that the metric reads as

$$\mathrm{d}s^2 = a^2(\tau)\left[-(1+2\Psi)\mathrm{d}\tau^2 + (1-2\Phi)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\right], \tag{3.7}$$

where we have set the usual notation $A = \Psi$ and $D = 2\Phi$ using the Bardeen potentials. In the absence of any extra source of anisotropic stress and for perfect fluids, general relativity tells us that both potentials should be equivalent $\Phi = \Psi$. Also, it is usual to redefine the scalar velocity potential of component (n) by

$$v_{i(n)} = -\partial_i v_{(\mathrm{n})}, \qquad \theta_{(\mathrm{n})} \equiv i\vec{k}\cdot\vec{v}_{(\mathrm{n})} \qquad \text{and} \qquad u^\mu_{(\mathrm{n})} = \frac{1}{a}\left(1-\Psi, v^i_{(\mathrm{n})}\right) . \tag{3.8}$$

For each component (n), we can define an equation of state w, two variables for the perturbations, the density contrast δ and the velocity potential θ already introduced, and the physical sound speed defined in the fluid rest frame as

$$w_{(\mathrm{n})} \equiv \frac{\bar{P}_{(\mathrm{n})}}{\bar{\rho}_{(\mathrm{n})}}, \qquad \delta_{(\mathrm{n})} \equiv \frac{\delta\rho_{(\mathrm{n})}}{\bar{\rho}_{(\mathrm{n})}}, \qquad \theta_{(\mathrm{n})} \equiv i\vec{k}\cdot\vec{v}_{(\mathrm{n})}, \qquad c^2_{\mathrm{s},(\mathrm{n})} \equiv \left.\frac{\delta P_{(\mathrm{n})}}{\delta\rho_{(\mathrm{n})}}\right|_{\mathrm{rf}} . \tag{3.9}$$

while in an arbitrary frame we have the following relation

$$\delta P_{(\mathrm{n})} = c^2_{\mathrm{s},(\mathrm{n})}\delta\rho_{(\mathrm{n})} - (c^2_{\mathrm{s},(\mathrm{n})} - c^2_{\mathrm{a},(\mathrm{n})})\bar{\rho}'_{(\mathrm{n})}v_{(\mathrm{n})} , \tag{3.10}$$

making use of the adiabatic sound speed c_a, defined as

$$c^2_{\mathrm{a},(\mathrm{n})} \equiv \frac{\bar{P}'_{(\mathrm{n})}}{\bar{\rho}'_{(\mathrm{n})}} = w_{(\mathrm{n})} + \frac{w'_{(\mathrm{n})}}{\bar{\rho}'_{(\mathrm{n})}/\bar{\rho}_{(\mathrm{n})}} \quad \xrightarrow{\text{non-interacting}} \quad w_{(\mathrm{n})} - \frac{w'_{(\mathrm{n})}}{3\mathcal{H}\left(1+w_{(\mathrm{n})}\right)}. \tag{3.11}$$

Having specified the metric, fluid variables and sound speed conventions, we can now write the evolution equations in the absence of interactions. In this case, each component conserves its energy-momentum tensor independently, $\nabla_\nu T^{\mu\nu}_{(n)} = 0$, which leads to the standard continuity and Euler equations governing the evolution of the density contrast and velocity divergence of each perfect fluid component. These equations provide the reference system for the rest of this section since the effects of interactions will be introduced right after by replacing the individual conservation law with a covariantly defined transfer four-vector $Q^\mu_{(n)}$. With the conventions established above, the scalar perturbation equations for a non-interacting perfect fluid in Newtonian gauge are[5] [7]

$$\delta'_{(\mathrm{n})} = 3\mathcal{H}(w_{(\mathrm{n})} - c^2_{\mathrm{s},(\mathrm{n})})\delta_{(\mathrm{n})} - (1+w_{(\mathrm{n})})\theta_{(\mathrm{n})} + 3(1+w_{(\mathrm{n})})\Phi' - 9\mathcal{H}^2(c^2_{\mathrm{s},(\mathrm{n})} - c^2_{\mathrm{a},(\mathrm{n})})(1+w_{(\mathrm{n})})\frac{\theta_{(\mathrm{n})}}{k^2}, \tag{3.12}$$

$$\theta'_{(\mathrm{n})} = -\mathcal{H}(1-3c^2_{\mathrm{s},(\mathrm{n})})\theta_{(\mathrm{n})} + \frac{k^2 c^2_{\mathrm{s},(\mathrm{n})}}{1+w_{(\mathrm{n})}}\delta_{(\mathrm{n})} + k^2\Psi. \tag{3.13}$$

B. Covariantisation of couplings

The couplings introduced at the level of the background energy density conservation equations are a useful phenomenological starting point. However, this description is not enough once perturbations are included, since a cosmological covariantly formulation is needed. We therefore introduce the interaction for each component (n) through

$$\nabla_\nu T^{\mu\nu}_{(\mathrm{n})} = Q^\mu_{(\mathrm{n})}, \qquad \sum_n Q^\mu_{(\mathrm{n})} = 0. \tag{3.14}$$

The second condition guarantees total energy-momentum conservation. For later convenience we decompose the source four-vector as [8–10]

$$Q^\mu_{(\mathrm{n})} = Q_{(\mathrm{n})}\hat{u}^\mu + F^\mu_{(\mathrm{n})}, \tag{3.15}$$

$$Q_{(\mathrm{n})} = \bar{Q}_{(\mathrm{n})} + \delta Q_{(\mathrm{n})}, \tag{3.16}$$

$$F^\mu_{(\mathrm{n})} = \frac{1}{a}(0, \vec{f}_{(\mathrm{n})}) = \frac{1}{a}(0, -\partial^j f_{(\mathrm{n})}), \tag{3.17}$$

---

[5] For baryons, the Euler equation has an additional term from Thomson scattering with photons of the form $+\frac{4\rho_\gamma}{3\rho_{(\mathrm{b})}}\, a n_e \sigma_\mathrm{T}\,(\theta_\gamma - \theta_{(\mathrm{b})})$, which accounts for the momentum exchange between the two fluids. Moreover, for baryons $w_{(\mathrm{b})} \simeq c^2_{\mathrm{a},(\mathrm{b})} \simeq 0$, while the sound speed $c^2_{\mathrm{s},(\mathrm{b})}$ is small but nonzero. The standard treatment (theoretically [7] and in codes [2–4]) retains only the pressure gradient term $k^2 c^2_{\mathrm{s},(\mathrm{b})}\delta_{(\mathrm{b})}$, whereas the terms $-3\mathcal{H}c^2_{\mathrm{s},(\mathrm{b})}\delta_{(\mathrm{b})}$, $-9\mathcal{H}^2 c^2_{\mathrm{s},(\mathrm{b})}\theta_{(\mathrm{b})}/k^2$ and $+3\mathcal{H}c^2_{\mathrm{s},(\mathrm{b})}\theta_{(\mathrm{b})}$ are usually neglected, being suppressed by $c^2_{\mathrm{s},(\mathrm{b})}$ with respect to the terms of order $\mathcal{H}$. In any case, the effect of neglecting them is extremely small and below any potential effect on any observable.

with $\hat{u}_\mu F^\mu_{(\mathrm{n})} = 0$. Here $\bar{Q}_{(\mathrm{n})}$ is the background energy-transfer rate, $\delta Q_{(\mathrm{n})}$ is its perturbation, and $f_{(\mathrm{n})}$ is the scalar momentum transfer potential. The vector $\hat{u}^\mu$ defines the frame in which the energy transfer is measured, it does not need to coincide with the four-velocity of any particular fluid and usually is called the reference frame. We write it following the same convection as before as

$$\hat{u}^\mu = \frac{1}{a}(1-\Psi, \hat{v}^i), \qquad \hat{\theta} \equiv i\vec{k}\cdot\hat{\vec{v}}. \tag{3.18}$$

At the level of components, the above decomposition of the interaction translated into

$$Q_{0\,(n)} = -a\left[(1+\Psi)\bar{Q}_{(\mathrm{n})} + \delta Q_{(\mathrm{n})}\right], \qquad Q_{j\,(n)} = -a\partial_j\left(\bar{Q}_{(\mathrm{n})}\hat{v} + f_{(\mathrm{n})}\right). \tag{3.19}$$

where the condition $\sum_n Q^\mu_{(\mathrm{n})} = 0$ implies

$$\sum_n \bar{Q}_{(\mathrm{n})} = 0, \qquad \sum_n \delta Q_{(\mathrm{n})} = 0, \qquad \sum_n f_{(\mathrm{n})} = 0, \tag{3.20}$$

that for two interacting fluids this reduces to $\bar{Q}_{(1)} = -\bar{Q}_{(2)}$, $\delta Q_{(1)} = -\delta Q_{(2)}$ and $f_{(1)} = -f_{(2)}$.

1. *Interacting background equations*

With the previous decomposition we have set all the rules for computing the relevant quantities and equations to describe both the background and perturbations evolution when any interaction between at least two components is proposed. Regarding the background, the energy density conservation law for each component now reads as

$$\bar{\rho}'_{(\mathrm{n})} + 3\mathcal{H}(1+w_{(\mathrm{n})})\bar{\rho}_{(\mathrm{n})} = a\bar{Q}_{(\mathrm{n})}. \tag{3.21}$$

Consequently, for an interacting component, we have the following relations of the physical and adiabatic sound speeds

$$\begin{aligned} c^2_{\mathrm{a},(\mathrm{n})} &= w_{(\mathrm{n})} + \frac{w'_{(\mathrm{n})}}{-3\mathcal{H}\left(1+w_{(\mathrm{n})}\right) + a\bar{Q}_{(\mathrm{n})}/\bar{\rho}_{(\mathrm{n})}} = w_{(\mathrm{n})} + \bar{\rho}_{(\mathrm{n})}\frac{w'_{(\mathrm{n})}}{\bar{\rho}'_{(\mathrm{n})}}, \\ c^2_{\mathrm{s},(\mathrm{n})} - c^2_{\mathrm{a},(\mathrm{n})} &= c^2_{\mathrm{s},(\mathrm{n})} - w_{(\mathrm{n})} - \frac{w'_{(\mathrm{n})}}{-3\mathcal{H}\left(1+w_{(\mathrm{n})}\right) + a\bar{Q}_{(\mathrm{n})}/\bar{\rho}_{(\mathrm{n})}}. \end{aligned} \tag{3.22}$$

2. *Interacting perturbation equations*

We now can also write the perturbation equations for a generic interacting component (n). We use the Newtonian gauge, the previous definitions and convention for the velocity variables and the interaction decomposition defined in Eq. (3.17) where the quantities $\bar{Q}_{(\mathrm{n})}$, $\delta Q_{(\mathrm{n})}$ and $f_{(\mathrm{n})}$ denote respectively the background energy transfer, its perturbation and the scalar momentum-transfer potential. The reference-frame velocity entering the interaction is denoted by $\hat{\theta}$ as commented above. Then, the density contrast and Euler equation for each component (n) are[6] [9, 11]

$$\begin{aligned} \delta'_{(\mathrm{n})} =& 3\mathcal{H}\big(w_{(\mathrm{n})} - c^2_{\mathrm{s},(\mathrm{n})}\big)\delta_{(\mathrm{n})} - (1+w_{(\mathrm{n})})\theta_{(\mathrm{n})} + 3(1+w_{(\mathrm{n})})\Phi' - 9\mathcal{H}^2\big(c^2_{\mathrm{s},(\mathrm{n})} - c^2_{\mathrm{a},(\mathrm{n})}\big)(1+w_{(\mathrm{n})})\frac{\theta_{(\mathrm{n})}}{k^2} \\ &+ \frac{a}{\rho_{(\mathrm{n})}}\delta Q_{(\mathrm{n})} + \frac{aQ_{(\mathrm{n})}}{\rho_{(\mathrm{n})}}\left[\Psi - \delta_{(\mathrm{n})} + 3\mathcal{H}\big(c^2_{\mathrm{s},(\mathrm{n})} - c^2_{\mathrm{a},(\mathrm{n})}\big)\frac{\theta_{(\mathrm{n})}}{k^2}\right], \end{aligned} \tag{3.23}$$

$$\begin{aligned} \theta'_{(\mathrm{n})} =& -\mathcal{H}\big(1 - 3c^2_{\mathrm{s},(\mathrm{n})}\big)\theta_{(\mathrm{n})} + \frac{k^2 c^2_{\mathrm{s},(\mathrm{n})}}{1+w_{(\mathrm{n})}}\delta_{(\mathrm{n})} + k^2\Psi \\ &+ \frac{aQ_{(\mathrm{n})}}{(1+w_{(\mathrm{n})})\rho_{(\mathrm{n})}}\left[\hat{\theta} - \big(1 + c^2_{\mathrm{s},(\mathrm{n})}\big)\theta_{(\mathrm{n})}\right] + \frac{ak^2}{(1+w_{(\mathrm{n})})\rho_{(\mathrm{n})}}f_{(\mathrm{n})}. \end{aligned} \tag{3.24}$$

With the chosen decomposition, $f_{(\mathrm{n})}$ enters only the Euler equation, while $\delta Q_{(\mathrm{n})}$ enters only the density equation. On the other hand, $\bar{Q}$ being a background quantity enters both equations.

[6] In the case of baryons, the Euler equation (3.24) must again be supplemented with the Thomson-scattering term. Again, since $w_{(\mathrm{b})} \simeq c^2_{\mathrm{a},(\mathrm{b})} \simeq 0$ the interaction terms proportional to these quantities vanish, while those proportional to $c^2_{\mathrm{s},(\mathrm{b})}$ do not. Whether the latter should be neglected, as is usually done for the analogous terms in the non-interacting case, is not clearly addressed in the literature since although they are extremely small, in principle all of them should be consistently taken into account.

3. Note on reference frame velocity and expansion

Consider an interaction whose background coupling can be written as $Q = \mathcal{I}H$, where $\mathcal{I}$ is any scalar. It is common practice in the literature to consider the Hubble as a function with no perturbations so it ends up covariantising the interaction such that $\delta Q = H\delta\mathcal{I}$. At the homogeneous level, the Hubble rate is unambiguous. Away, that means perturbations, however, the symbol H alone does not define a unique local covariant quantity so a covariant completion is required. For that, we need to specify a timelike congruence u^μ, which at linear level is nothing more than specifying a velocity reference frame, and replace the Hubble function by the expansion scalar $\Theta \equiv \nabla_\mu u^\mu$. Therefore, the common prescription in which H is kept unperturbed is a phenomenological choice, but it is not by itself a fully defined covariant interaction. Although it is certainly enough before the era of precision cosmology, neglecting the perturbation of the local expansion might in principle have some impact with the precision of current and, particularly, future observations. Since SymBoltz solves the complete Einstein-Boltzmann system without any approximation-switching schemes at all, we avoid introducing this additional approximation here following the philosophy of the code. Then, $\Theta = \nabla_\mu u^\mu$, which for the background is nothing more than $3H$, is [8, 10]

$$\Theta = \nabla_\mu u^\mu = \partial_\mu u^\mu + \Gamma^\mu_\mu\lambda u^\lambda = \frac{3}{a}\left[\mathcal{H}(1-\Psi) - \Phi' + \frac{\theta}{3}\right] . \tag{3.25}$$

When we considered the decomposition of any coupling Q^μ, we introduced a vector $\hat{u}^\mu$ which defined the reference frame in which the energy transfer was measured. In principle, we did not decide what would be that reference frame. In this project and in the search for the most natural and general one, we use all the components to define the inertial density to weight it so that that reference frame is set by the total velocity. In our notation that means:

$$\hat{\Theta} \equiv \Theta_{\text{tot}} = \nabla_\mu \hat{u}^\mu \implies \hat{v}^i = \frac{\sum_{(\text{n})}(1+w_{(\text{n})})\bar{\rho}_{(\text{n})}v^i_{(\text{n})}}{\sum_{(\text{n})}(1+w_{(\text{n})})\bar{\rho}_{(\text{n})}} \text{ or equivalently } \hat{\theta} = \frac{\sum_{(\text{n})}(1+w_{(\text{n})})\bar{\rho}_{(\text{n})}\theta_{(\text{n})}}{\sum_{(\text{n})}(1+w_{(\text{n})})\bar{\rho}_{(\text{n})}} \tag{3.26}$$

We use the $\hat{}$ to denote that total weighted velocity related variables from now on. Other choices are possible, in fact any other other choice of the congruence is perfectly acceptable and possible while use consistently. Given any choice, we end up with the following:

$$\bar{Q} = \frac{\bar{\mathcal{I}}\bar{\Theta}}{3}, \qquad \delta Q = \frac{\bar{\Theta}}{3}\delta\mathcal{I} + \frac{\bar{\mathcal{I}}}{3}\delta\Theta, \tag{3.27}$$

where, for each choice, $\bar{\Theta}$ and $\delta\Theta$ would represent that case, following a corresponding new definition of equation (3.26). However, as said, here we stay with the the case of the weighted total velocity.

IV. IMPLEMENTATION AND VALIDATION OF SYMBOLTZ FOR INTERACTING MODELS

In this section we show how interacting cosmological models can be implemented in SymBoltz and later we use several different cases as tests to validate SymBoltz architecture for interacting scenarios by comparing its outputs to the ones of the well established Boltzmann code CLASS. For doing the validation we want to demonstrate that SymBoltz reproduces the background and perturbation evolution obtained with the same level of precision as CLASS. Apart from that, we we want to show that implementing a new interaction in SymBoltz, no matter how elaborated may be or even when the interaction modifies several fluids or has a non-linear dependence on the densities, is considerably more direct than in previous approaches. Only three function will be needed to implemented, following the decomposition explained in previous Section III B.

A. Implementation

Once the user has the 4-vector Q^μ_{ij} for the interaction between fluids (i,j), it can be decomposed into $\bar{Q}_{ij}$, δQ_{ij} and f_{ij} following Section III B and implemented in SymBoltz.

We modify the minimal ΛCDM example in the code documentation, where every variable, parameter and equation of the entire standard model is specified in around 300 lines of code. This example is straightforward to modify with arbitrary interaction equations. To not duplicate the entire model for every interaction, we wrap this in a function that takes arbitrary interaction expressions as input. Our resulting model is found here[7], and primarily makes the following modifications to the standard ΛCDM model:

[7] https://hersle.github.io/SymBoltz.jl/stable/interacting/

```
function interacting_model(; name = :QΛCDM,
    Qbc = 0, QbΛ = 0, QcΛ = 0, δQbc = 0, δQbΛ = 0, δQcΛ = 0, fQbc = 0, fQbΛ = 0, fQcΛ = 0
)

    pars = @parameters begin
        αbc, αcΛ, αbΛ
        ...
    end

    vars = @variables begin
        ρb(t), [backwards = true],
        ρc(t), [backwards = true],
        ρΛ(t), [backwards = true],
        Qb(t), Qc(t), QΛ(t),
        δQb(t,k), δQc(t,k), δQΛ(t,k),
        fQb(t,k), fQc(t,k), fQΛ(t,k), θ(t,k),
        ...
    end

    ...

    eqs = [
        ...

        Qb ~ -Qbc + QbΛ
        Qc ~ +Qbc + QcΛ
        QΛ ~ -QcΛ - QbΛ
        δQb ~ -δQbc + δQbΛ
        δQc ~ +δQbc + δQcΛ
        δQΛ ~ -δQcΛ - δQbΛ
        fQb ~ -fQbc + fQbΛ
        fQc ~ +fQbc + fQcΛ
        fQΛ ~ -fQcΛ - fQbΛ

        Dτ(ρb) ~ -3ℋ*(1+wb)*ρb + a*Qb
        Dτ(ρc) ~ -3ℋ*(1+wc)*ρc + a*Qc
        Dτ(ρΛ) ~ -3ℋ*(1+wΛ)*ρΛ + a*QΛ
        θ ~ ((ρΛ+PΛ)*θΛ + (ρh+Ph)*θh + (ρν+Pν)*θν + (ρc+Pc)*θc + (ργ+Pγ)*θγ + (ρb+Pb)*θb) / ((ρΛ+PΛ) + (ρh+Ph) +
        ↪  (ρν+Pν) + (ρc+Pc) + (ργ+Pγ) + (ρb+Pb))

        Dτ(δb) ~ -θb + (a * Qb / ρb) * (Ψ - δb + 3ℋ*csb2*θb/k^2) + (a * δQb / ρb) + 3*Dτ(Φ)
        Dτ(θb) ~ -ℋ*θb + k^2*csb2*δb + k^2*Ψ + (a * Qb / ρb) * (θ - θb*(1+csb2)) + (a * k^2 / ρb) * fQb -
        ↪  4/3*Dτ(κ)*ργ/ρb*(θγ-θb)
        Dτ(δc) ~ -θc + (a * Qc / ρc) * (Ψ - δc) + (a * δQc / ρc) + 3*Dτ(Φ)
        Dτ(θc) ~ -ℋ*θc + k^2*Ψ + (a * Qc / ρc) * (θ - θc) + (a * k^2 / ρc) * fQc
        Dτ(δΛ) ~ -(1+wΛ)*θΛ - 3ℋ*(cΛs2-wΛ)*δΛ - 9*(ℋ/k)^2*(1+wΛ)*(cΛs2-cΛa2)*θΛ + (a * QΛ / ρΛ) * (Ψ - δΛ +
        ↪  3ℋ*(cΛs2-cΛa2)*θΛ/k^2) + (a * δQΛ / ρΛ) + 3*(1+wΛ)*Dτ(Φ)
        Dτ(θΛ) ~ -ℋ*(1-3*cΛs2)*θΛ + cΛs2/(1+wΛ)*k^2*δΛ + k^2*Ψ  + (a * QΛ / (ρΛ*(1+wΛ))) * (θ - θΛ*(1+cΛs2)) + (a *
        ↪  k^2 / (ρΛ*(1+wΛ))) * fQΛ
    ]

    ...
end
```

These are straightforward translations of the continuity and Euler equations in Section III B. We adopt the following sign convention for Q^μ (and for $\bar{Q}$, δQ and f) to satisfy condition 3.20:

$$Q^\mu_{(b)} = -Q^\mu_{bc} + Q^\mu_{b\Lambda}, \tag{4.1}$$

$$Q^\mu_{(c)} = \quad Q^\mu_{bc} + Q^\mu_{c\Lambda}, \tag{4.2}$$

$$Q^\mu_{(\Lambda)} = -Q^\mu_{b\Lambda} - Q^\mu_{c\Lambda}. \tag{4.3}$$

Here `...` stands for unmodified lines from the ΛCDM model, and `[backwards = true]` tells SymBoltz to integrate the background continuity equations for $\{\rho_b, \rho_c, \rho_\Lambda\}$ from their densities today and backwards in time (see Section II). The Unicode characters make the code correspond one-to-one with our equations. Due to the approximation-free nature of the code, it is not necessary to implement tight-coupling approximations.

The `interacting_model()` function can now be used as a constructor for building models with arbitrary $Q^\mu_{(b)}$, $Q^\mu_{(c)}$ and $Q^\mu_{(\Lambda)}$ by passing any symbolic expressions to the keyword arguments. We first create a non-interacting model `M` and `@unpack` its symbolic variables that we want to access. Then we can create several instances of `interacting_model()` with arbitrary interaction expressions. For example, to conveniently construct the model in Eq. 4.7:

```
M = interacting_model()
@unpack a, ℋ, ρc, ρΛ, δc, δΛ, θ, θc, θΛ, Ψ, Φ, k, αcΛ = M

M1 = interacting_model(
    QcΛ = αcΛ * ℋ * ρΛ / a,
    δQcΛ = αcΛ / a * (ℋ * (ρΛ*δΛ - ρΛ*Ψ) + ρΛ * (θ/3 - Dτ(Φ))),
    fQcΛ = αcΛ * ℋ * ρΛ / (a*k^2) * (θc - θ),
    name = :αCDM,
)
```

Other interacting models are constructed in exactly the same way. This makes it easy to create many different interacting models.

B. Validation

Having described how the interactions are implemented, we now test the precision and flexibility of SymBoltz. We compare its predictions with the corresponding implementations in CLASS, considering first standard cosmological models without interactions, then pure momentum transfer models, and finally models with both energy and momentum exchange (see Ref. [12] for a recent review on interactions). This allows us to check separately the standard non-interacting evolution, modifications affecting only the perturbations, and cases in which the interaction also changes the background evolution. The comparison is performed using the background densities and Hubble rate, the matter power spectrum, and the TT and EE CMB angular power spectra. For the following plots we are going to fix the cosmological parameters to certain values, although the results are similar for any other choice of parameters. In particular, we fix the reduced Hubble constant to $h = 0.70$, the today dark matter density to $\Omega_\mathrm{c} = 0.25$, the today baryons density to $\Omega_\mathrm{b} = 0.05$, Helium fraction to $Y_\mathrm{He} = 0.25$, the amplitude of scalar perturbations to $A_\mathrm{s} = 2 \cdot 10^-9$, the scalar spectral index to $n_\mathrm{s} = 0.94$, while we fix to have two massless neutrinos and one massive neutrino with total mass $\sum_i m_\nu = 0.02$ eV. Any other parameter is fixed to the default value.

1. *Reference cases: no interactions*

Here we study the behaviour of SymBoltz in comparison with CLASS for the reference cases ΛCDM, wCDM and the Chevallier–Polarski–Linder (CPL) parametrization [13, 14] also known as w_0w_aCDM. The first and most important result is that, in the absence of interactions, SymBoltz reproduces the standard and very typical cosmological cases as expected. In particular in Figures 2 and 3 we display and compare the results for ΛCDM (black), two wCDM cases with $w = -0.9$ (red) and $w = -1.1$ (blue), and two w_0w_aCDM cases $(w_0, w_a) = (-0.9, +0.2)$ (green) and $(w_0, w_a) = (-1.1, -0.2)$ (gold). For the background evolution (upper panel), the cold dark matter density ρ_c, baryon density ρ_b, dark energy/cosmological constant density ρ_Λ and Hubble rate $H(a)$ from SymBoltz follow the corresponding CLASS curves for all five cases. The curves are basically on top of each other in the upper panels for all the different dark energy models considered here the relative difference with respect to CLASS, defined as $\log_{10}\left|\frac{X_\mathrm{SymBoltz} - X_\mathrm{CLASS}}{X_\mathrm{CLASS}}\right|$ remains well below the 0.001%. (lower panel). The same result is found for the matter power spectrum (middle panel) since the differences to CLASS remain for all $k \in [10^{-4}, 1]$ h/Mpc well below the 1%. Finally, the cosmic microwave background spectra, in this case TT and EE, also agree well below the 1% difference with CLASS except at very first low multipoles in the EE spectrum, where the spectrum approaches zero. The resulting spike in the relative difference is therefore caused by the normalization to a vanishingly small CLASS value, rather than by a significant absolute discrepancy between the two codes.

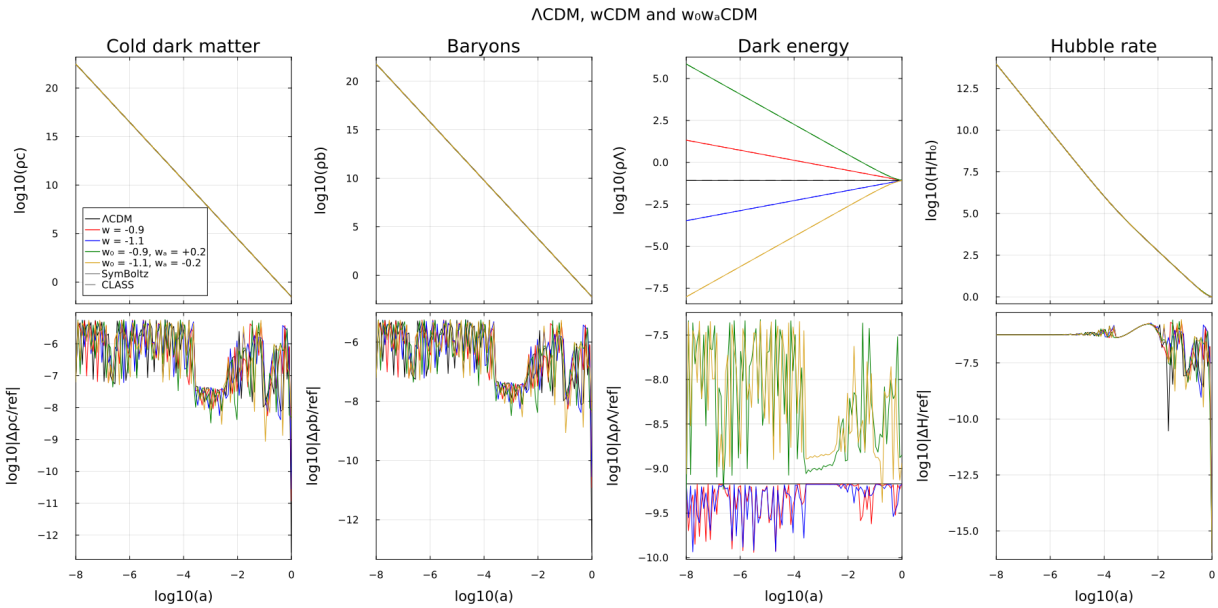


FIG. 2. Background comparison between CLASS and SymBoltz for ΛCDM (black), wCDM with $w = -0.9$ (red) and $w = -1.1$ (blue), and $w_0 w_a$CDM with $(w_0, w_a) = (-0.9, +0.2)$ (green) and $(w_0, w_a) = (-1.1, -0.2)$ (gold). The upper panels show the background evolution, while the lower panels show the relative difference, calculated as $\log_{10} |(X_{\mathrm{SymBoltz}} - X_{\mathrm{CLASS}})/X_{\mathrm{CLASS}}|$.

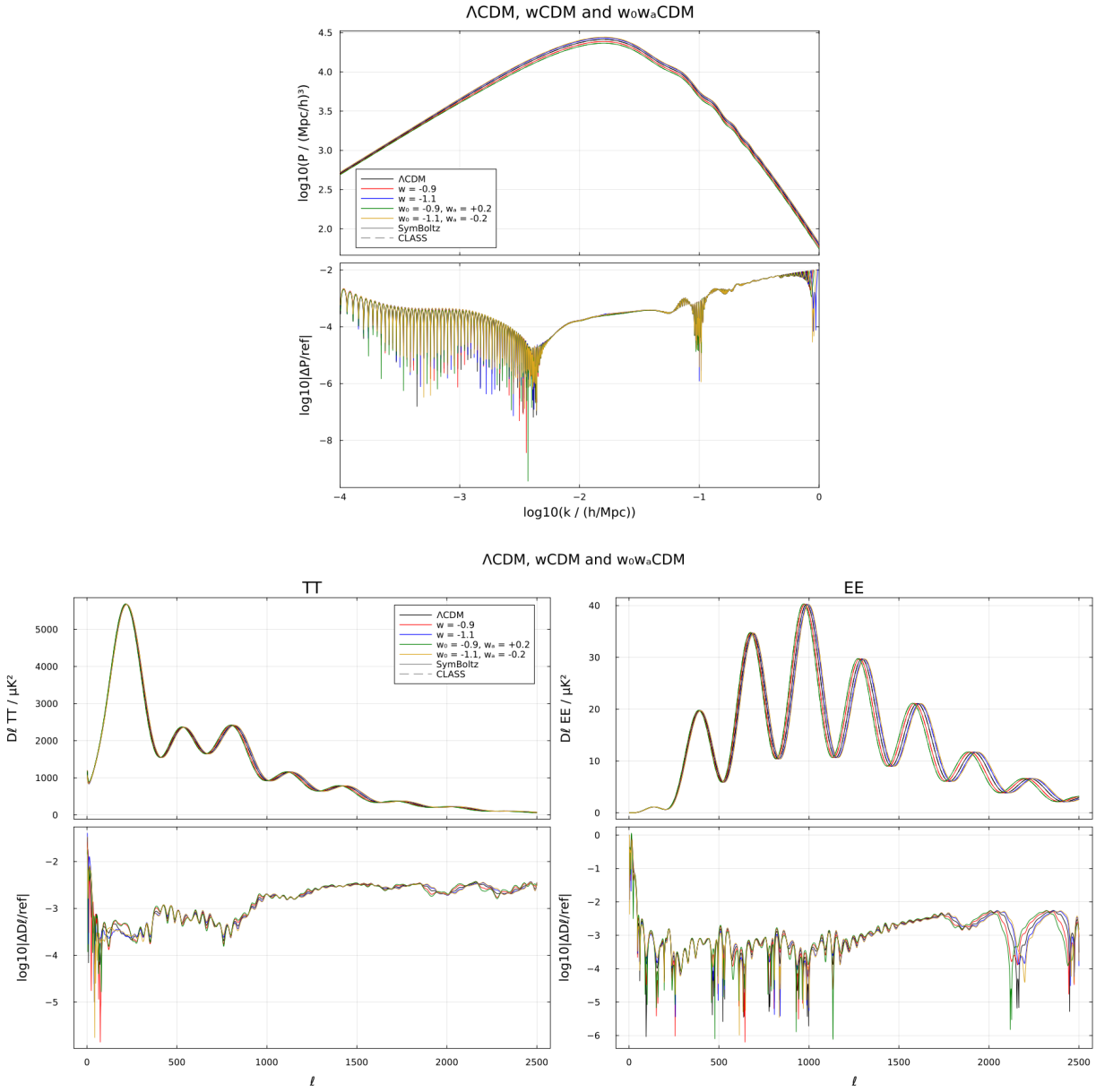


FIG. 3. Perturbation comparison between CLASS and SymBoltz for ΛCDM (black), wCDM with $w = -0.9$ (red) and $w = -1.1$ (blue), and w_0w_aCDM with $(w_0, w_a) = (-0.9, +0.2)$ (green) and $(w_0, w_a) = (-1.1, -0.2)$ (gold). Top: matter power spectrum $P(k)$. Bottom: CMB TT and EE angular power spectra. The lower panels show the relative difference, calculated as $\log_{10} |(X_{\rm SymBoltz} - X_{\rm CLASS})\,/X_{\rm CLASS}|$.

2. *Pure momentum transfer*

Now we consider interactions to be implemented into SymBoltz. In particular, we begin with the most simple case one can imagine: the trilogy of momentum transfer models. Here a coupling between two species appears in the form of a pure momentum coupling which preserves the background evolution by the virtue of the cosmological principle while coupling the velocities of both coupled species. They do not only preserve background equations unchanged but also density contrast equations remain formally unaltered. Then, the one and only modification appears on the Euler equations giving thus the name of pure momentum transfer. One of the first appearances of these models dates back to Ref. [15], but a pleiad of models around the original one have emerged [16–30]. In our case, we are going to use as example of an implementation in SymBoltz the so-called αCDM model [31], the βCDM model [32] and the γCDM model [33] given we have direct access to their corresponding CLASS implementation, which will allow us to test the capabilities of SymBoltz.

2.a) *Dark energy – dark matter*

A pure momentum interaction that acts between dark matter and dark energy and that was first studied in Ref. [31] and lately extended in Ref. [23], while a public CLASS repository is available here. The interaction[8] is defined by $Q^\mu = \bar{\alpha}(u^\mu_\Lambda - u^\mu_c)$, so that in the decomposition of Section III B translates to:

$$\begin{aligned} \bar{Q} &= 0 \\ \delta Q &= 0 \\ f &= \tfrac{\bar{\alpha}}{k^2}(\theta_\Lambda - \theta_c) \qquad \text{with} \quad \bar{\alpha} = \alpha \tfrac{3H_0^3}{8\pi G} = \alpha H_0 \rho_{cr} \end{aligned} \tag{4.4}$$

In Figures 4 and 5, we display as before the comparison with respect to CLASS. Given the background evolution is unaltered by the interaction, we obtain similar results as in the reference non-interacting cases with differences well below the 0.001% level. Looking at the perturbations, which are modified by the interaction, we see in the middle plot that for a large range of possible coupling parameters α, the differences with respect to CLASS are well below the 1% at all scales. Similar results can be obtained from the TT and EE cosmic microwave background (lower panel) with differences well below the 1% level except in a few low-ℓ EE modes, something that already happened for the reference non interacting case and was explained there. Consequently, the SymBoltz implementation demonstrate perfect match with previous codes.

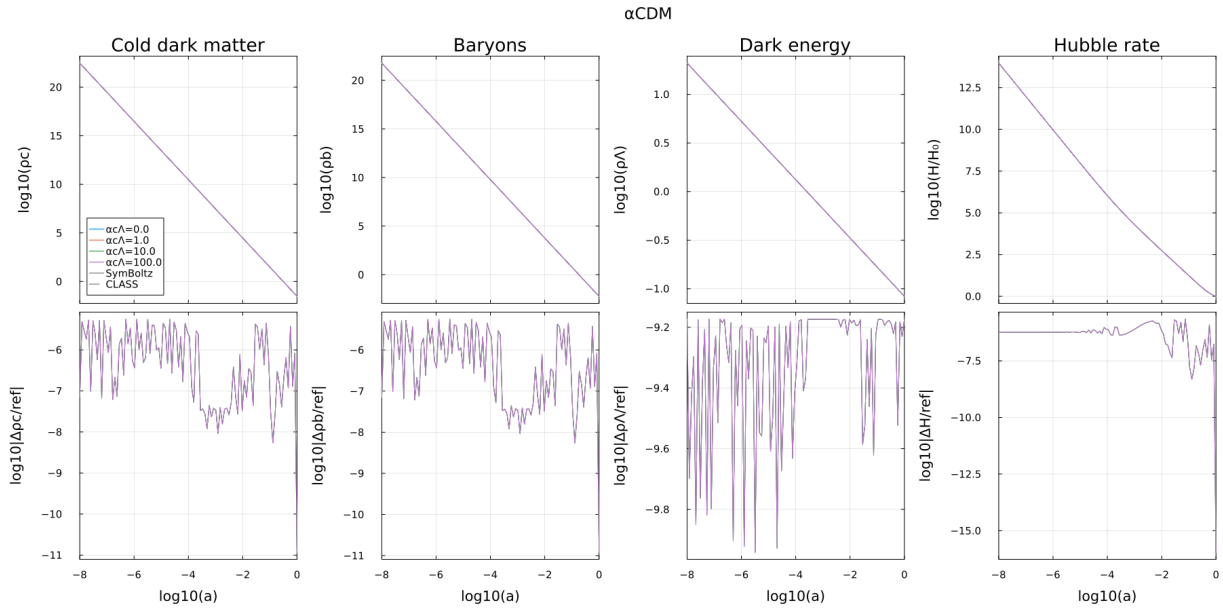


FIG. 4. Background comparison between CLASS and SymBoltz for αCDM. The upper panels show the background evolution, while the lower panels show the relative difference, calculated as $\log_{10}|(X_{\text{SymBoltz}} - X_{\text{CLASS}})/X_{\text{CLASS}}|$.

[8] In the code the coupling parameter follows always the notation αij where i and j are the interacting species in each case. Here, to ease the notation, we simply use α for this interaction (β and γ for the other pure momentum cases) which will correspond in SymBoltz to $\alpha \equiv \alpha c\Lambda$.

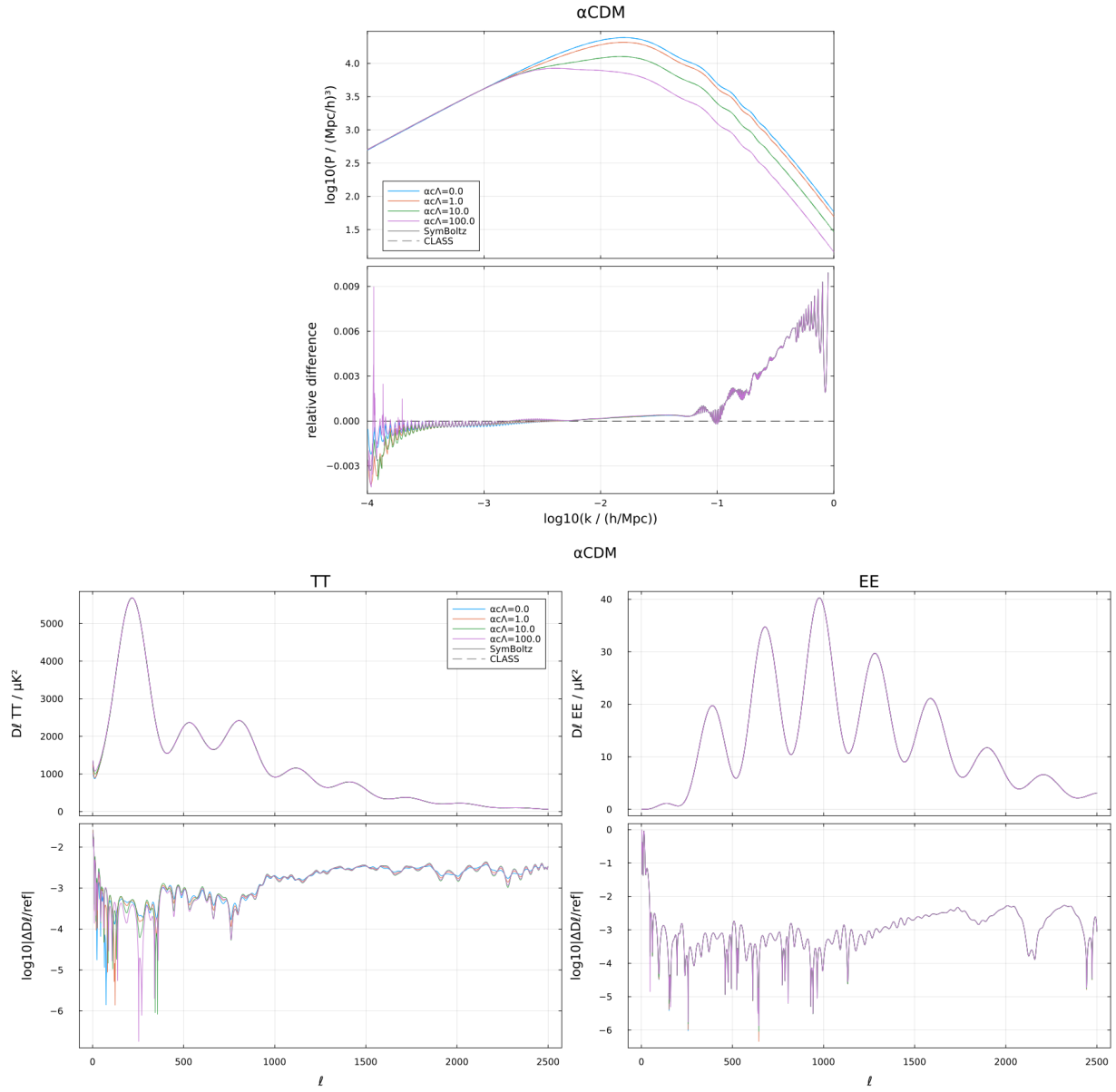


FIG. 5. Perturbation comparison between CLASS and SymBoltz for αCDM. Top: matter power spectrum $P(k)$. Bottom: CMB TT and EE angular power spectra. The lower panels show the relative difference, calculated as $\log_{10}|(X_{\rm SymBoltz} - X_{\rm CLASS})/X_{\rm CLASS}|$.

2.b) Dark energy – baryons

A pure momentum interaction but now between baryons and dark energy which was first studied in [32] inspired by Ref. [34]. The interaction is defined by $Q^\mu = \bar{\beta}(u^\mu_\Lambda - u^\mu_{\rm b})$ so that we have in this case the following decomposition:

$$\begin{aligned} \bar{Q} &= 0 \\ \delta Q &= 0 \\ f &= \tfrac{\bar{\beta}}{k^2}(\theta_\Lambda - \theta_{\rm b}) \qquad \text{with} \quad \bar{\beta} = \beta \tfrac{3H_0^3}{8\pi G} = \beta H_0 \rho_{\rm cr} \end{aligned} \tag{4.5}$$

In Figure 6 we display the matter power spectrum (top) and the TT and EE cosmic microwave background (bottom) while we omit the background corresponding plot for the background given the modification is not altered due to the interaction and we showed above how the SymBoltz results are indistinguishable from the CLASS ones. Also, regarding the matter power spectrum and the cosmic microwave background, the comparison to CLASS demonstrates SymBoltz results with less than 1% difference, again.

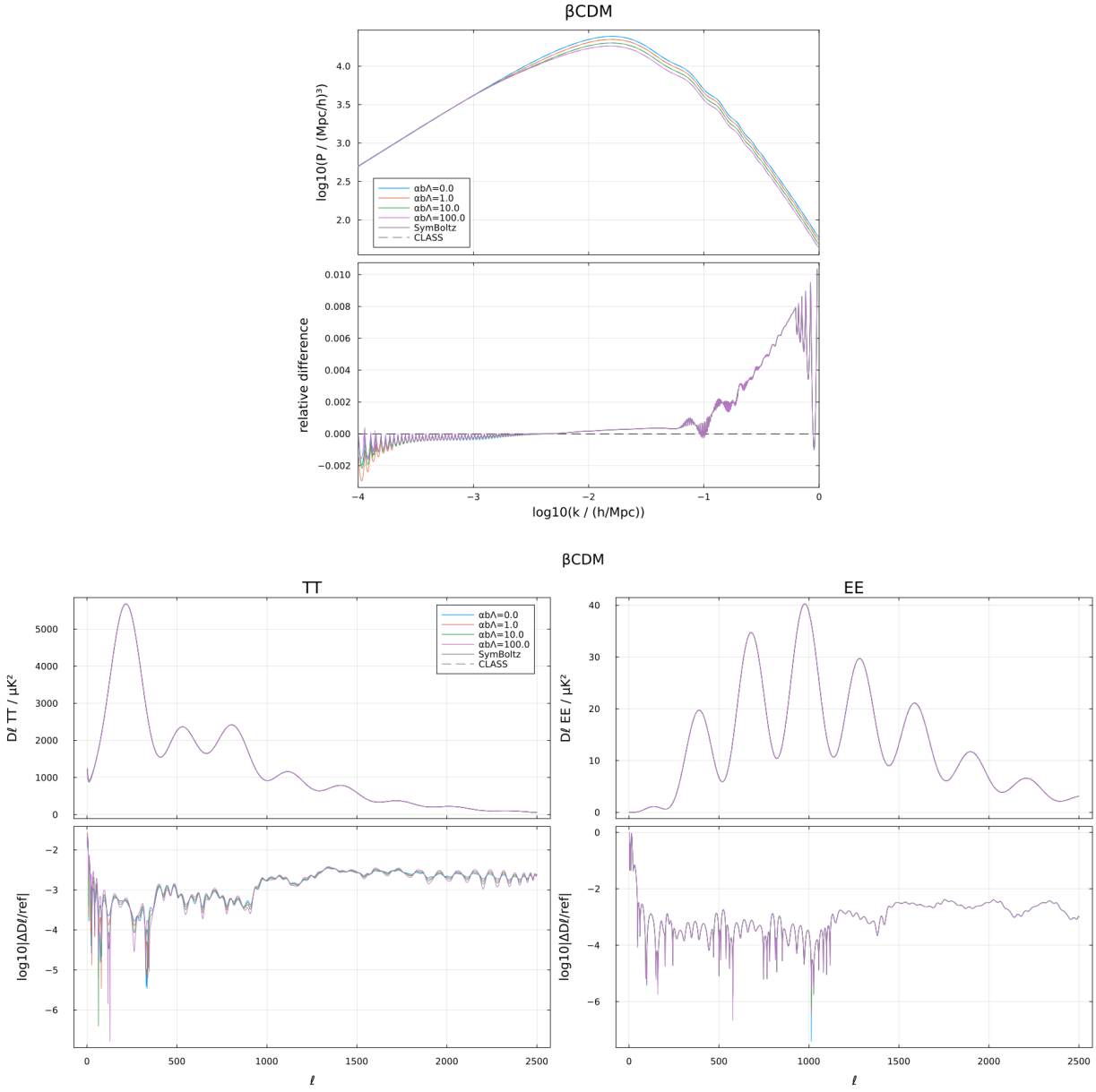


FIG. 6. βCDM comparison between CLASS and SymBoltz. Top: matter power spectrum $P(k)$. Bottom: CMB TT and EE.

2.c) Dark matter – baryons

Finally a pure momentum interaction between dark matter and baryons first studied in Ref. [33] and given by $Q^\mu = \frac{\gamma}{\Omega_{\rm b}\rho_{cr,0}}(T^{\alpha\beta}u^b_\alpha u^b_\beta)^2(u^\mu_{\rm b} - u^\mu_{\rm c})$. Again, here we omit the background comparison since the interaction cannot alter it. On the other hand, when we look at the matter power spectrum in Figure 7, we find that for linear scales differences remain again well below the 1% level between SymBoltz and CLASS. If we move towards smaller scales, $k > 0.1$ h/Mpc, where a non-linear prescription or a N-body analysis will be needed and for very large values of the coupling parameter some percentage differences can appear in that extreme case. On the other hand, the cosmic microwave background remains well below the 1% with similar behaviour as previous momentum transfer remaining independent of the value of the coupling parameter.

$$\begin{aligned} \bar{Q} &= 0 \\ \delta Q &= 0 \\ f &= -\gamma\frac{\Omega_{\rm b,0}\rho_{cr,0}}{a^6k^2}(\theta_{\rm c} - \theta_{\rm b}) \end{aligned} \tag{4.6}$$

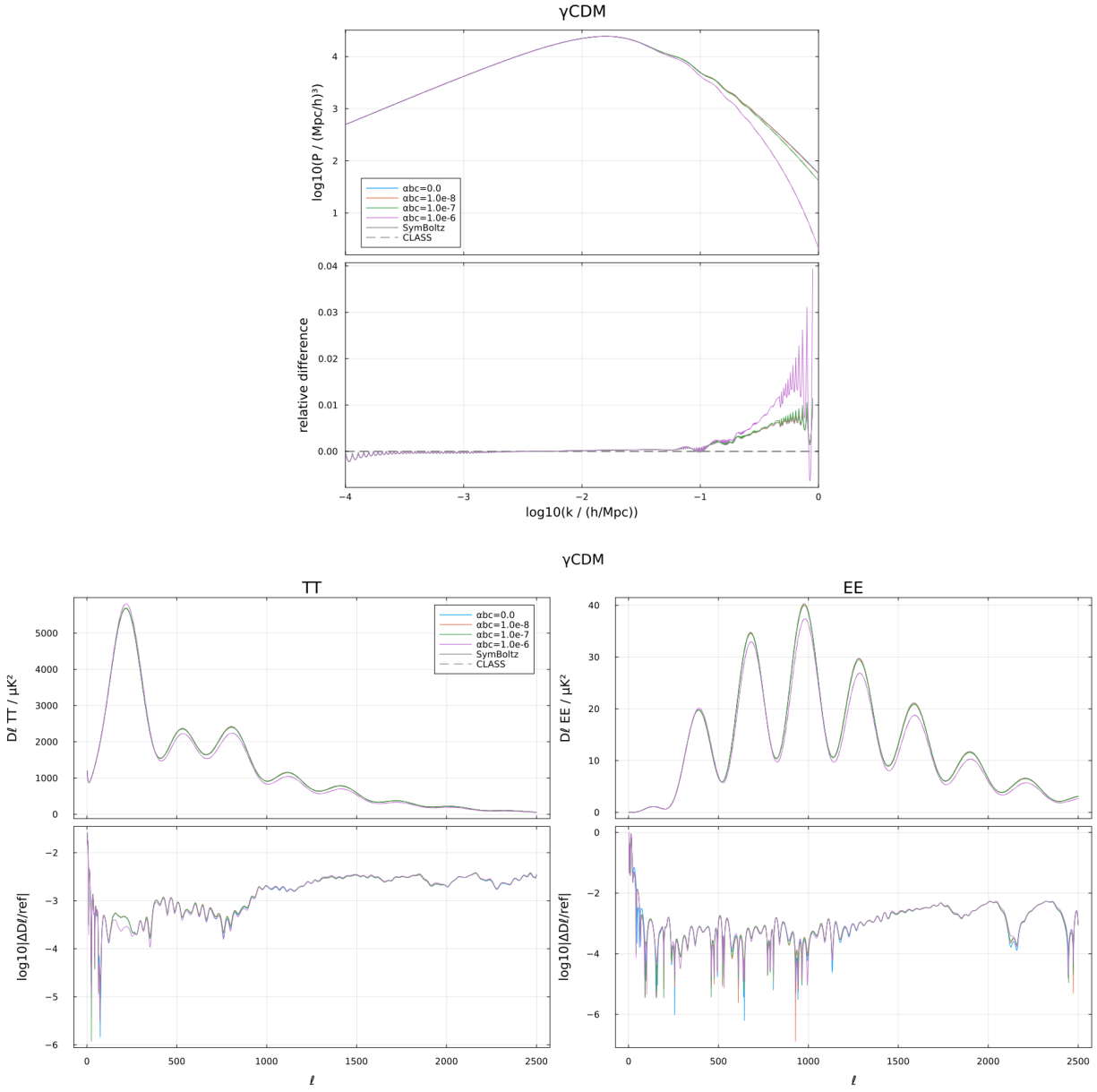


FIG. 7. γCDM comparison between CLASS and SymBoltz. Top: matter power spectrum $P(k)$. Bottom: CMB TT and EE.

3. Dark energy – dark matter: energy and momentum transfer

Previous interactions involved only a pure momentum exchanged and kept the background evolution of the interacting species as in standard non-interacting cases, ΛCDM or wCDM models. Now we can extend the testing of the capabilities of SymBoltz to scenarios when energy transfer appears and, thus, background evolution is altered. This will also serve as a test of the backwards integration capabilities of SymBoltz version 2.0.0 explained at Section II, since solving the energy conservation equations with $\bar{Q} \neq 0$ requires either shooting or backwards integration for solving $\bar{\rho}'_{(\mathrm{n})} + 3\mathcal{H}(1 + w_{(\mathrm{n})})\bar{\rho}_{(\mathrm{n})} = a\bar{Q}_{(\mathrm{n})}$. Here, we use as example cases for testing the capabilities of SymBoltz a linear dark energy - dark matter interaction with energy and momentum transfer, a non linear dark energy - dark matter interaction with energy and momentum transfer,

3.a) Linear coupling

We consider a linear coupling between dark energy and dark matter (several examples can be found in Refs. [10,

35–39] defined[9] by the coupling $Q^\nu_{\rm c} = \zeta \frac{\hat{\Theta}}{3} T^{\alpha\beta}_\Lambda u^\Lambda_\alpha u^\Lambda_\beta u^\nu_{\rm c}$, which translates to the following decomposition:

$$\begin{aligned} \bar{Q} &= \zeta \frac{\mathcal{H}}{a} \rho_\Lambda \\ \delta Q &= \zeta \frac{\mathcal{H}}{a} \rho_\Lambda (\delta_\Lambda - \Psi) + \zeta \frac{\rho_\Lambda}{a} (\frac{\hat{\theta}}{3} - \Phi') \\ f &= \zeta \frac{\mathcal{H}}{k^2 a} \rho_\Lambda (\theta_{\rm c} - \hat{\theta}) \end{aligned} \tag{4.7}$$

In Figures 8 and 9 we display the background evolution of the different energy densities of the interacting and non-interacting components (top panel), the matter power spectrum (middle panel) and the cosmic microwave background (bottom panel) for several values of the coupling parameter ζ of the interaction while we also display the comparison to the same model in each case using CLASS code. Regarding the background evolution, we can see how both codes perfectly match up to a difference well below the 0.001% level in the densities of the interacting components, ρ_Λ and $\rho_{\rm c}$. Such difference lower than 0.001% is comparable to the relative difference on the non-interacting component, $\rho_{\rm b}$, between SymBoltz and CLASS. Also when we consider the Hubble function, we obtain the difference between codes to be, again, on the 0.001% level. Consequently, even when we have the background evolution altered by an interaction, we have demonstrated how both codes are completely equivalent. Considering the perturbation, both codes show perfect agreement in their respective matter power spectrum for almost all scales and for the TT or EE cosmic microwave background for all multipoles. Even for unrealistic values of the coupling, like $\zeta = 1$ which have a extremely suppressed matter power spectrum and a cosmic microwave background incompatible with Planck, both codes have differences in their output below the 1% level. The only difference in the codes appear in the large scale matter power spectrum. This is due to the interaction since it is absence in the case with $\zeta = 0$. Also, matter power spectrum coming from SymBoltz is perfectly smooth while the one from CLASS shows some irregularities. Then the cause is expected to come from CLASS and can be cause by [**DF**: To explain]... . Consequential, we have demonstrated how SymBoltz can handle interacting models as precise as other Boltzmann codes, while have a more user-friendly way of implementing them by just writing three lines of code with the details of the interaction and also having no approximation or switching-schemes.

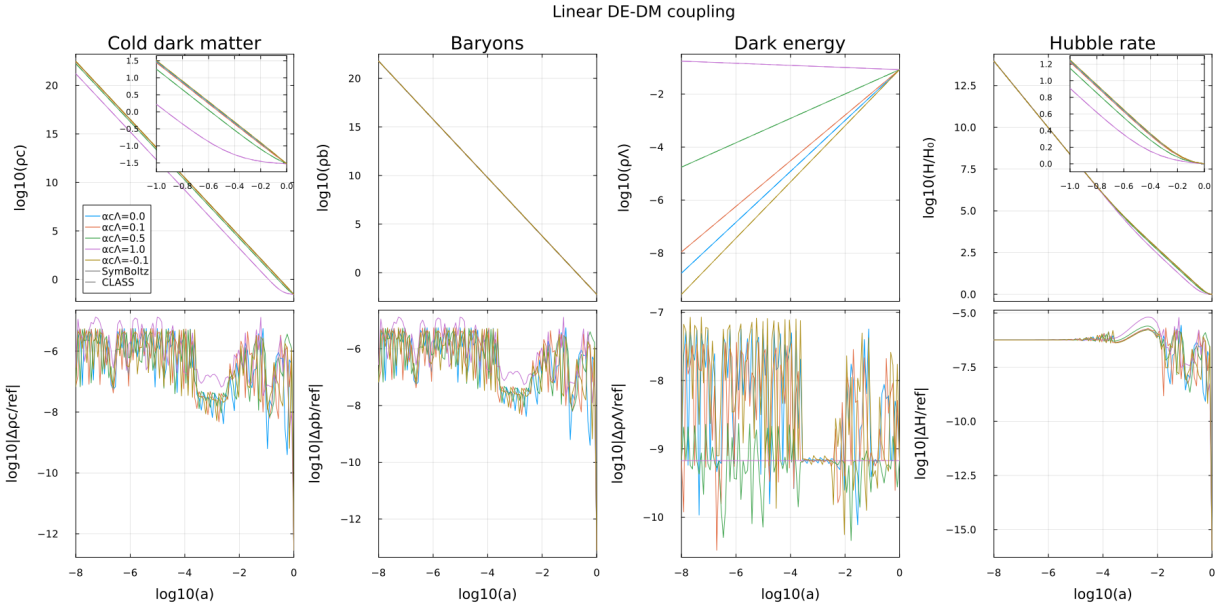


FIG. 8. Background comparison between CLASS and SymBoltz for the linear DE-DM interaction $Q^\nu_{\rm c} = \zeta \frac{\hat{\Theta}}{3} T^{\alpha\beta}_\Lambda u^\Lambda_\alpha u^\Lambda_\beta u^\nu_{\rm c}$. The upper panels show the background evolution, while the lower panels show the relative difference, calculated as $\log_{10} |(X_{\rm SymBoltz} - X_{\rm CLASS}) / X_{\rm CLASS}|$.

[9] In the code the coupling parameter always follows the notation αij where i and j are the interacting species in each case. Here, to ease the notation, we simply use ζ for this interaction which will correspond in SymBoltz to $\zeta \equiv \alpha c\Lambda$.

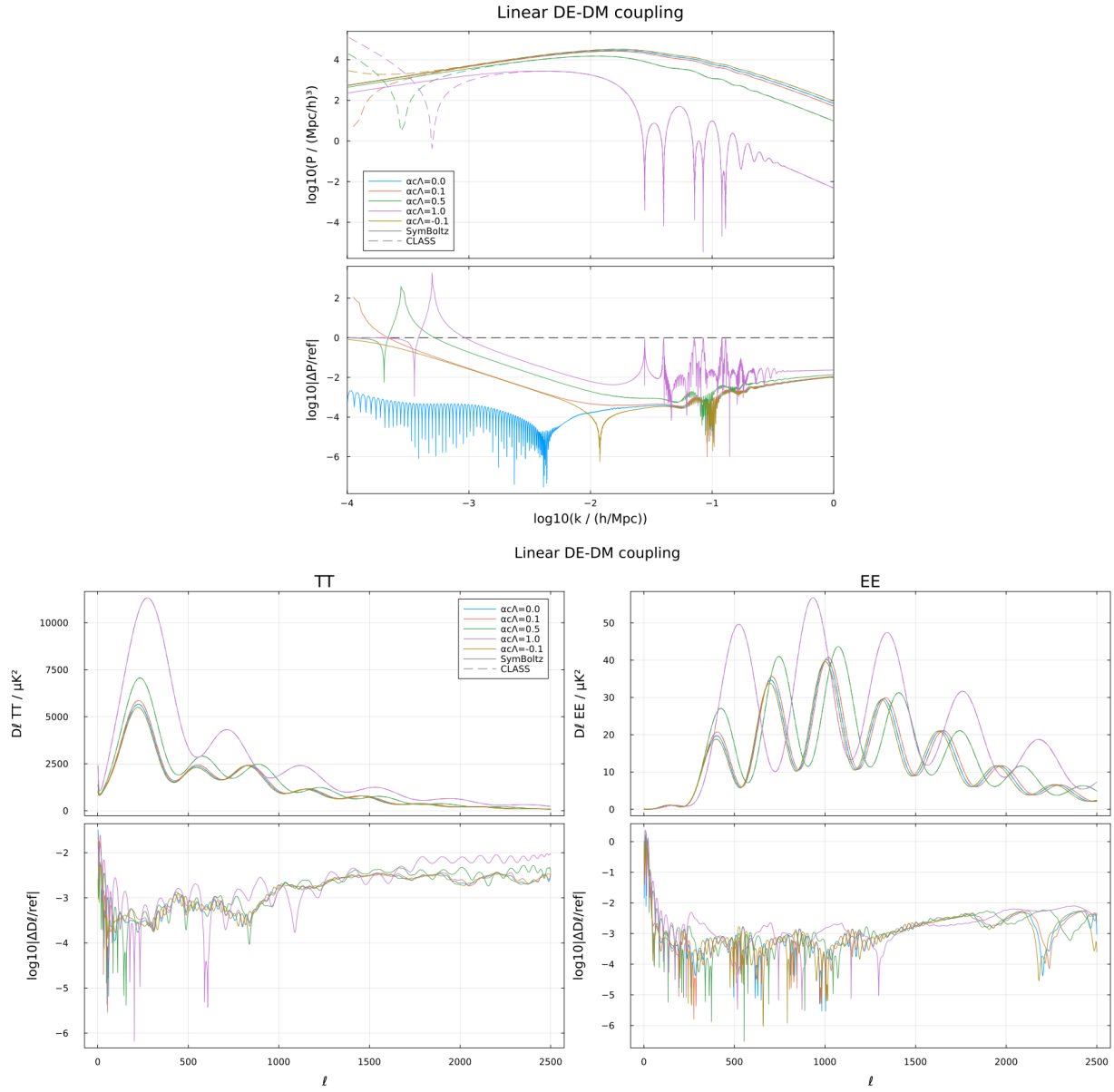


FIG. 9. Perturbation comparison between CLASS and SymBoltz for the linear DE-DM interaction $Q^\nu_{\rm c} = \zeta \frac{\hat{\Theta}}{3} T^{\alpha\beta}_{\Lambda} u^{\Lambda}_{\alpha} u^{\Lambda}_{\beta} u^{\nu}_{\rm c}$. Top: matter power spectrum $P(k)$. Bottom: CMB TT and EE angular power spectra. The lower panels show the relative difference, calculated as $\log_{10} |(X_{\rm SymBoltz} - X_{\rm CLASS})/X_{\rm CLASS}|$.

3.b) Non-linear coupling

We consider a non linear coupling between dark energy and dark matter (several examples can be found in Refs. [40–42]) defined by the 4-vector $Q^\mu_{\rm c} = \zeta \hat{\Theta} \frac{\left(T^{\alpha\beta}_{({\rm c})} u^{({\rm c})}_{\alpha} u^{({\rm c})}_{\beta}\right)\left(T^{\gamma\delta}_{(\Lambda)} u^{(\Lambda)}_{\gamma} u^{(\Lambda)} \delta\right)}{T^{\alpha\beta}_{({\rm c})} u^{({\rm c})}_{\alpha} u^{({\rm c})}_{\beta} + T^{\gamma\delta}_{(\Lambda)} u^{(\Lambda)}_{\gamma} u^{(\Lambda)}_{\delta}} u^{\mu}_{\rm c}$ while, in order to implement it into SymBoltz, we decompose it as

$$\begin{aligned} \bar{Q} &= 3\zeta \frac{\mathcal{H}}{a} \frac{\rho_{\rm c}\rho_\Lambda}{\rho_{\rm c}+\rho_\Lambda} \\ \delta Q &= 3\zeta \frac{\mathcal{H}}{a} \frac{\rho_{\rm c}\rho_\Lambda}{\rho_{\rm c}+\rho_\Lambda} \left[\frac{\rho_\Lambda \delta_{\rm c} + \rho_{\rm c}\delta_\Lambda}{\rho_{\rm c}+\rho_\Lambda} - \Psi\right] + \zeta \frac{1}{a} \frac{\rho_{\rm c}\rho_\Lambda}{\rho_{\rm c}+\rho_\Lambda} \left(\hat{\theta} - 3\Phi'\right) \\ f &= 3\zeta \frac{\mathcal{H}}{k^2 a} \frac{\rho_{\rm c}\rho_\Lambda}{\rho_{\rm c}+\rho_\Lambda} \left(\theta_{\rm c} - \hat{\theta}\right) \end{aligned} \tag{4.8}$$

In Figures 10 and 11 we display the background evolution of the different energy densities of the interacting and

non-interacting components (top panel), the matter power spectrum (middle panel) and the cosmic microwave background (bottom panel) for several values of the coupling parameter ζ of the interaction while we also display the comparison to the same model in each case using CLASS code. Although the modifications produced by this interaction are phenomenologically similar to those obtained in the linear case, studying this model provides a non-trivial additional test. The energy-transfer rate now depends non-linearly on the background densities, and both codes must consistently evaluate this non-linear dependence and its corresponding first-order perturbation throughout the cosmological evolution. Then, regarding the background evolution, we can see how again both codes perfectly match up to a difference well below the 0.001% level in the densities of the interacting components. Also, when looking into the Hubble function, we obtain the difference between codes to be the same. Considering the perturbations, again both codes show perfect agreement in their respective matter power spectrum for almost all scales and for the TT or EE cosmic microwave background for all multipoles. As before, the only differences in the codes appear in the very low-ℓ EE cosmic microwave background and in the large scale matter power spectrum. Both were explained before and have now the same origin. Consequential, we have demonstrated how SymBoltz can handle non-linear interacting models as precise as other Boltzmann codes.

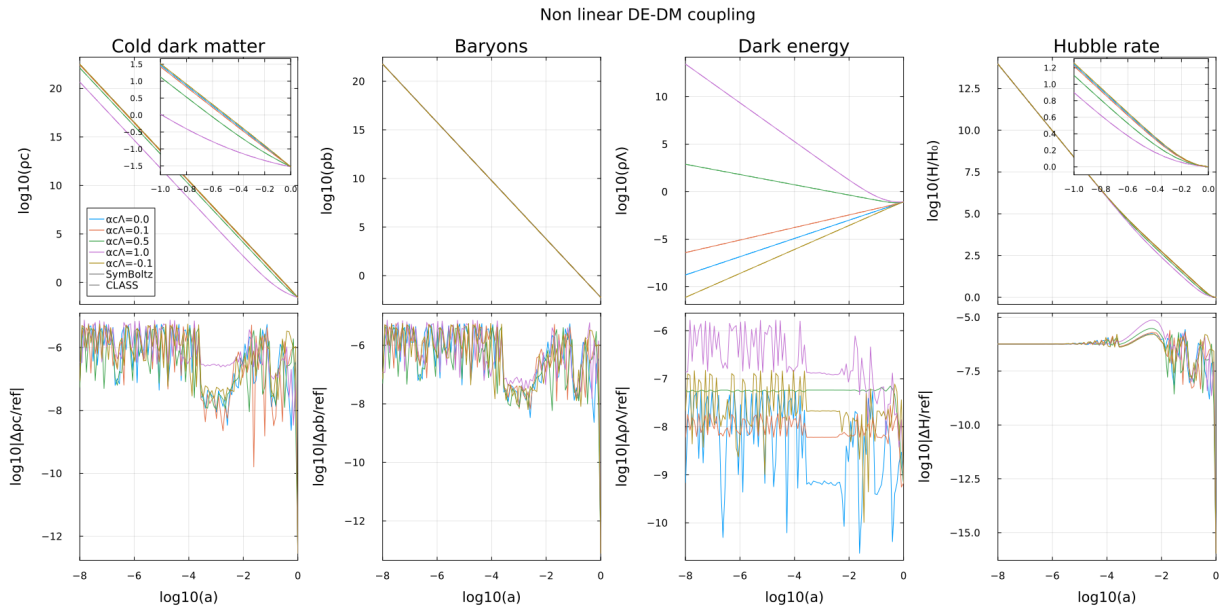


FIG. 10. Background comparison between CLASS and SymBoltz for the non-linear DE-DM interaction. The upper panels show the background evolution, while the lower panels show the relative difference, calculated as $\log_{10}|(X_{\mathrm{SymBoltz}} - X_{\mathrm{CLASS}})/X_{\mathrm{CLASS}}|$.

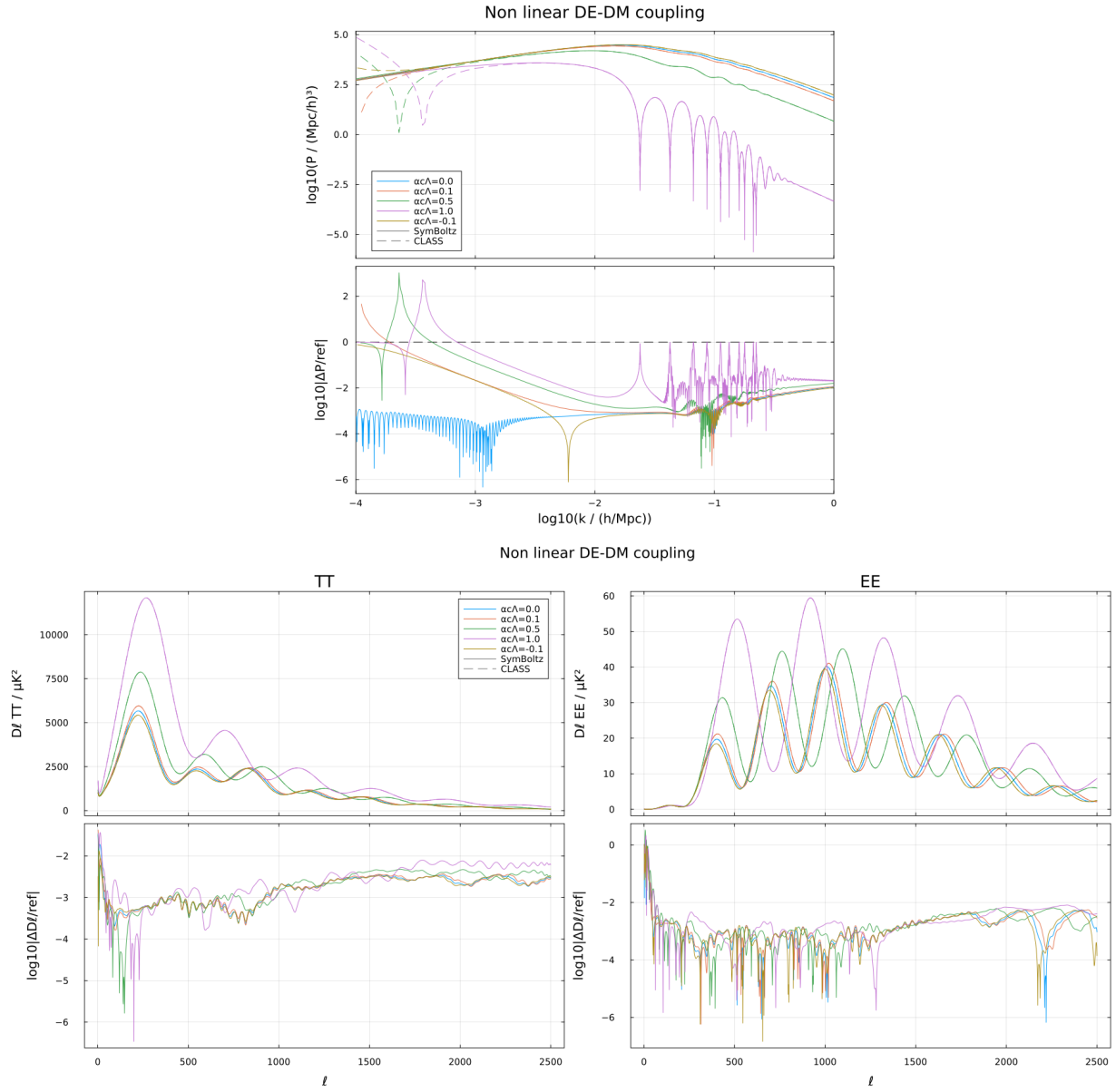


FIG. 11. Perturbation comparison between CLASS and SymBoltz for the non-linear DE-DM interaction. Top: matter power spectrum $P(k)$. Bottom: CMB TT and EE angular power spectra. The lower panels show the relative difference, calculated as $\log_{10}|(X_{\mathrm{SymBoltz}} - X_{\mathrm{CLASS}})/X_{\mathrm{CLASS}}|$.

The results presented above establish SymBoltz as a precise and powerful framework for studying interacting cosmologies, while also making it simple to modify and extend. Any new interaction can be implemented by specifying only the three quantities Q, δQ and f, each of which is obtained directly from the chosen interaction model. In practice, incorporating a new interaction amounts to adding a single line containing these three quantities. Its agreement with CLASS for standard non-interacting cosmologies, pure momentum transfer models, and both linear and non-linear energy-momentum transfer models demonstrates that the code accurately reproduces the background and perturbation evolution across a broad range of physically distinct scenarios. This accuracy is achieved while SymBoltz solves the full stiff Einstein-Boltzmann system using implicit methods, without relying on approximation switching schemes. This absence of approximation switching schemes is particularly important when an interaction modifies a component that uses an scheme, like baryons, since in conventional Boltzmann codes the interaction must also be consistently propagated through the corresponding approximation regimes and switching conditions. Most importantly, this capability is preserved when the cosmological model is modified: a new interaction is introduced directly through the three quantities Q, δQ and f, which are incorporated into the same general system of equations. Thus, SymBoltz combines the precision required for cosmological predictions with a flexible and modular structure that makes the exploration of general interacting models direct, systematic, and considerably simpler than modifying

a conventional Boltzmann code.

V. ALL TOGETHER AT THE SAME TIME: SEVERAL INTERACTIONS

Consider this system for multiple interactions by couples between dark energy, dark matter and baryons:

$$\nabla_\nu T^{\mu\nu}_{(\mathrm{n})} = Q^\mu_{(\mathrm{n})}, \qquad \sum_n Q^\mu_{(\mathrm{n})} = 0. \tag{5.1}$$

In order to study a real example to highlight the simplicity of SymBoltz when dealing with modifications, we have to make a choice of the interactions to study, although SymBoltz allows for any interaction no matter how complicated it would be. In our case we use the following ones at the same time all between 1) baryons and dark matter, 2) baryons and dark energy and 3) dark matter and dark energy of the form[10]:

$$Q^\nu_{bc} = \alpha_{bc}\frac{\hat\Theta}{3}\left[T^{\alpha\beta}_{(b)}u^{(b)}_\alpha u^{(b)}_\beta + T^{\alpha\beta}_{(c)}u^{(c)}_\alpha u^{(c)}_\beta\right]u^\nu_{\mathrm{c}} \tag{5.2}$$

$$Q^\nu_{b\Lambda} = \alpha_{b\Lambda}\frac{\hat\Theta}{3}\left[T^{\alpha\beta}_{(b)}u^{(b)}_\alpha u^{(b)}_\beta + T^{\alpha\beta}_{(\Lambda)}u^{(\Lambda)}_\alpha u^{(\Lambda)}_\beta\right]u^\nu_{\mathrm{b}} \tag{5.3}$$

$$Q^\nu_{c\Lambda} = \alpha_{c\Lambda}\frac{\hat\Theta}{3}\left[T^{\alpha\beta}_{(c)}u^{(c)}_\alpha u^{(c)}_\beta + T^{\alpha\beta}_{(\Lambda)}u^{(\Lambda)}_\alpha u^{(\Lambda)}_\beta\right]u^\nu_{\mathrm{c}} \tag{5.4}$$

Since we need to satisfy equations (5.1), we need to make a choice on a convection to ensure the covariant conservation of each of the coupled of stress energy tensor, or in other words to ensure $\sum_n Q^\mu_{(\mathrm{n})} = 0$. In our case we make the following choice:

$$\begin{aligned} Q^\nu_{(b)} &= -Q^\nu_{bc} + Q^\nu_{b\Lambda}, \\ Q^\nu_{(c)} &= +Q^\nu_{bc} + Q^\nu_{c\Lambda}, \\ Q^\nu_{(\Lambda)} &= -Q^\nu_{b\Lambda} - Q^\nu_{c\Lambda}. \end{aligned} \tag{5.5}$$

Having specified the three pairwise covariant transfer vectors and the sign convention used to combine them into the total exchange vectors $Q^\nu_{(b)}$, $Q^\nu_{(c)}$, and $Q^\nu_{(\Lambda)}$, we can now identify the three components of the general decomposition introduced in Section III B for each of the interaction that we are going to implement at the same time as

$\bar{Q}_{ij}$	δQ_{ij}	f_{ij}
$\bar{Q}_{bc} = \alpha_{bc}\frac{\mathcal{H}}{a}\left(\bar\rho_{\mathrm{b}} + \bar\rho_{\mathrm{c}}\right)$	$\delta Q_{bc} = \alpha_{bc}\frac{\mathcal{H}}{a}\left[\bar\rho_{\mathrm{b}}\delta_{\mathrm{b}} + \bar\rho_{\mathrm{c}}\delta_{\mathrm{c}} - \left(\bar\rho_{\mathrm{b}} + \bar\rho_{\mathrm{c}}\right)\Psi\right] + \alpha_{bc}\frac{\bar\rho_{\mathrm{b}} + \bar\rho_{\mathrm{c}}}{a}\left(\frac{\hat\theta}{3} - \Phi'\right)$	$f_{bc} = \alpha_{bc}\frac{\mathcal{H}}{ak^2}\left(\bar\rho_{\mathrm{b}} + \bar\rho_{\mathrm{c}}\right)\left(\theta_{\mathrm{c}} - \hat\theta\right)$
$\bar{Q}_{b\Lambda} = \alpha_{b\Lambda}\frac{\mathcal{H}}{a}\left(\bar\rho_{\mathrm{b}} + \bar\rho_{\Lambda}\right)$	$\delta Q_{b\Lambda} = \alpha_{b\Lambda}\frac{\mathcal{H}}{a}\left[\bar\rho_{\mathrm{b}}\delta_{\mathrm{b}} + \bar\rho_{\Lambda}\delta_{\Lambda} - \left(\bar\rho_{\mathrm{b}} + \bar\rho_{\Lambda}\right)\Psi\right] + \alpha_{b\Lambda}\frac{\bar\rho_{\mathrm{b}} + \bar\rho_{\Lambda}}{a}\left(\frac{\hat\theta}{3} - \Phi'\right)$	$f_{b\Lambda} = \alpha_{b\Lambda}\frac{\mathcal{H}}{ak^2}\left(\bar\rho_{\mathrm{b}} + \bar\rho_{\Lambda}\right)\left(\theta_{\mathrm{b}} - \hat\theta\right)$
$\bar{Q}_{c\Lambda} = \alpha_{c\Lambda}\frac{\mathcal{H}}{a}\left(\bar\rho_{\mathrm{c}} + \bar\rho_{\Lambda}\right)$	$\delta Q_{c\Lambda} = \alpha_{c\Lambda}\frac{\mathcal{H}}{a}\left[\bar\rho_{\mathrm{c}}\delta_{\mathrm{c}} + \bar\rho_{\Lambda}\delta_{\Lambda} - \left(\bar\rho_{\mathrm{c}} + \bar\rho_{\Lambda}\right)\Psi\right] + \alpha_{c\Lambda}\frac{\bar\rho_{\mathrm{c}} + \bar\rho_{\Lambda}}{a}\left(\frac{\hat\theta}{3} - \Phi'\right)$	$f_{c\Lambda} = \alpha_{c\Lambda}\frac{\mathcal{H}}{ak^2}\left(\bar\rho_{\mathrm{c}} + \bar\rho_{\Lambda}\right)\left(\theta_{\mathrm{c}} - \hat\theta\right)$

(5.6)

[10] As previously comment, the label Λ denotes dark energy of any form or a cosmological constant. Although an abuse of notation, we use it to keep as close as possible to the internal notation of SymBoltz.

This sign convection defined in eqs 5.5 for the three interactions, their components of the decomposition of Section III B is

$$\bar{Q}_{\rm b} = -\bar{Q}_{bc} + \bar{Q}_{b\Lambda}, \qquad \delta Q_{\rm b} = -\delta Q_{bc} + \delta Q_{b\Lambda}, \qquad f_{\rm b} = -f_{bc} + f_{b\Lambda}, \tag{5.7}$$

$$\bar{Q}_{\rm c} = +\bar{Q}_{bc} + \bar{Q}_{c\Lambda}, \qquad \delta Q_{\rm c} = +\delta Q_{bc} + \delta Q_{c\Lambda}, \qquad f_{\rm c} = +f_{bc} + f_{c\Lambda}, \tag{5.8}$$

$$\bar{Q}_{\Lambda} = -\bar{Q}_{b\Lambda} - \bar{Q}_{c\Lambda}, \qquad \delta Q_{\Lambda} = -\delta Q_{b\Lambda} - \delta Q_{c\Lambda}, \qquad f_{\Lambda} = -f_{b\Lambda} - f_{c\Lambda}. \tag{5.9}$$

These relations guarantee the cancellation of the energy and momentum transfers when all components are added together. Therefore, both at background and linear-perturbation level, the total stress-energy tensor is covariantly conserved for the interacting components as

$$\sum_{n\in\{b,c,\Lambda\}} Q^{\mu}_{\rm (n)} = 0 \qquad \Longrightarrow \qquad \nabla_{\nu}\left(T^{\mu\nu}_{(b)} + T^{\mu\nu}_{(c)} + T^{\mu\nu}_{(\Lambda)}\right) = 0. \tag{5.10}$$

This extremely complicated system is implemented into SymBoltz in the very same simple way as all the previous test cases. By just writing each of the Q, δQ and f of Table 5.6 into `interacting_model()` (see Section IV A), SymBoltz automatically creates the correct background and perturbations equations with no other requirement for the user. Then, the user only needs to specify the values of the coupling parameters $\alpha_{\rm ij}$ desired and the different output possibilities will be calculated. In particular for the interactions in Table 5.6 taken as proof of concept of the utility of SymBoltz, the user will just need to introduce the following:

```
...

M = interacting_model()
@unpack a, ℋ, ρb, ρc, ρΛ, δb, δc, δΛ, θ, θb, θc, θΛ, Ψ, Φ, k, αbc, αbΛ, αcΛ = M

# Background energy transfers
Qbc = αbc * ℋ / a * (ρb + ρc)
QbΛ = αbΛ * ℋ / a * (ρb + ρΛ)
QcΛ = αcΛ * ℋ / a * (ρc + ρΛ)

# Perturbations of the energy transfers
δQbc = αbc * ℋ / a * (ρb * δb + ρc * δc - (ρb + ρc) * Ψ) + αbc * (ρb + ρc) / a * (θ / 3 - Dτ(Φ))
δQbΛ = αbΛ * ℋ / a * (ρb * δb + ρΛ * δΛ - (ρb + ρΛ) * Ψ) + αbΛ * (ρb + ρΛ) / a * (θ / 3 - Dτ(Φ))
δQcΛ = αcΛ * ℋ / a * (ρc * δc + ρΛ * δΛ - (ρc + ρΛ) * Ψ) + αcΛ * (ρc + ρΛ) / a * (θ / 3 - Dτ(Φ))

# Scalar momentum-transfer potentials
fQbc = αbc * ℋ / (a * k^2) * (ρb + ρc) * (θc - θ)
fQbΛ = αbΛ * ℋ / (a * k^2) * (ρb + ρΛ) * (θb - θ)
fQcΛ = αcΛ * ℋ / (a * k^2) * (ρc + ρΛ) * (θc - θ)

M1 = interacting_model(; Qbc, QbΛ, QcΛ, δQbc, δQbΛ, δQcΛ, fQbc, fQbΛ, fQcΛ)
```

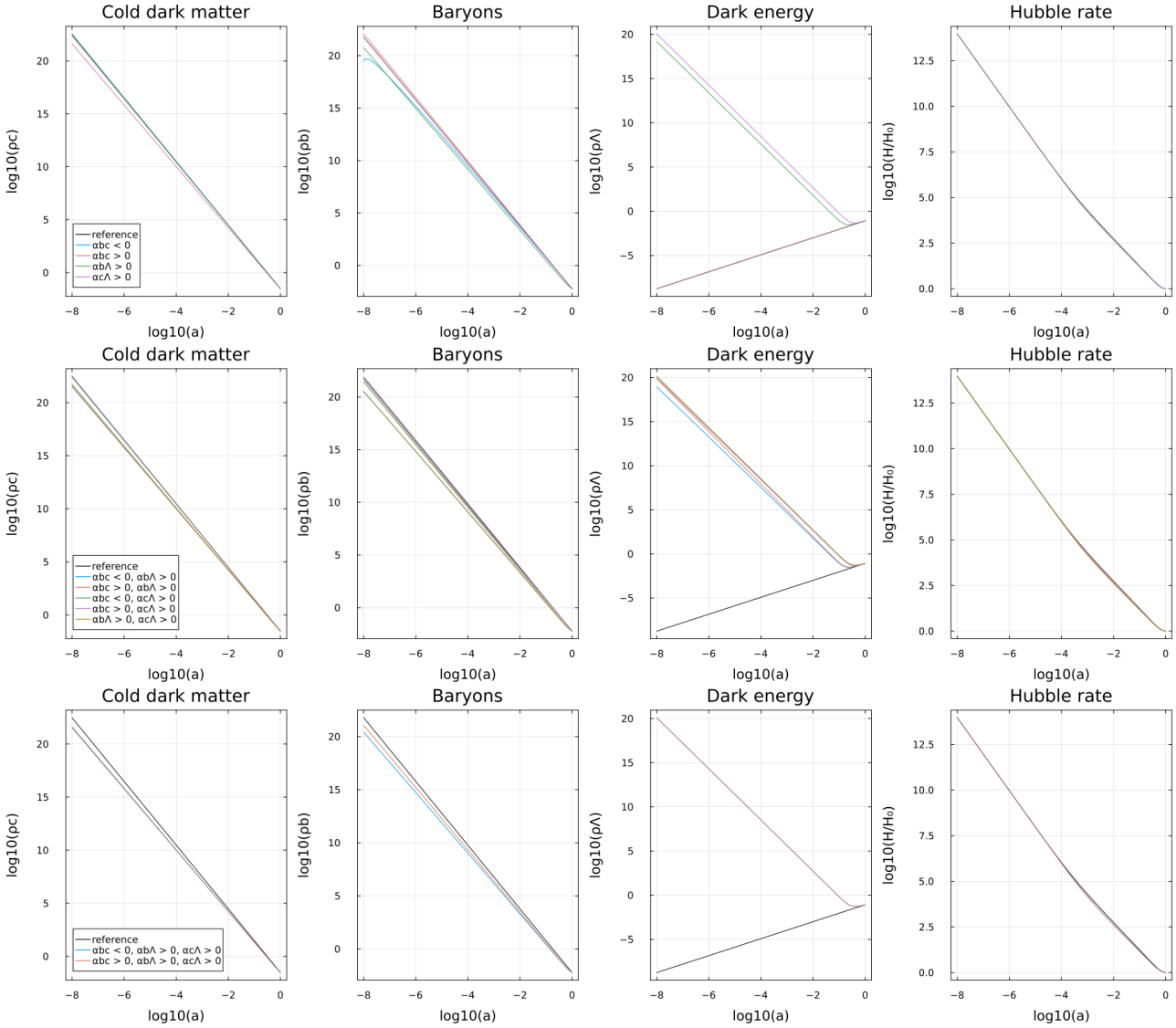


FIG. 12. Background evolution for the multi-interacting baryon-dark matter-dark energy model. From top to bottom, we show cases with one individual interaction, two simultaneous interactions, and the three interactions active at the same time.

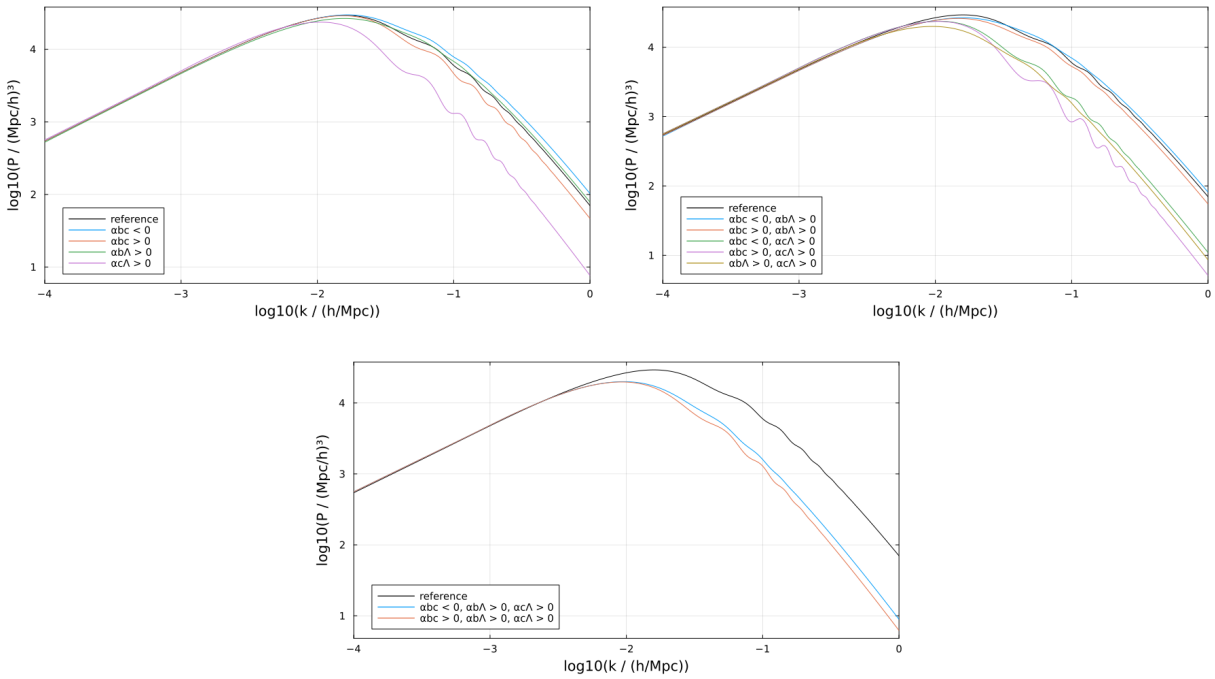


FIG. 13. Matter power spectrum for the multi-interacting baryon-dark matter-dark energy model. From top to bottom, we show cases with one individual interaction, two simultaneous interactions, and the three interactions active at the same time.

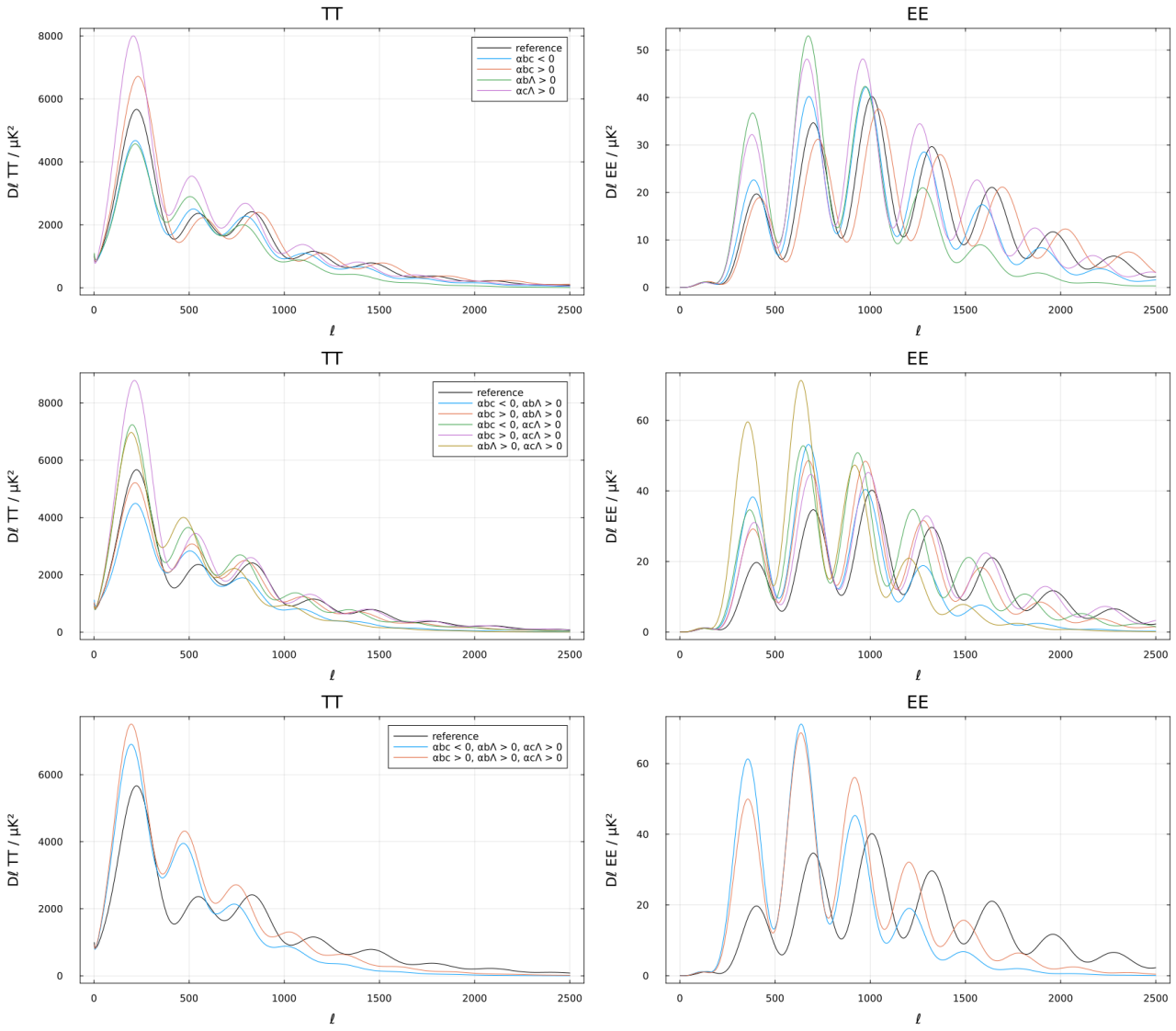


FIG. 14. CMB TT and EE angular power spectra for the multi-interacting baryon-dark matter-dark energy model. From top to bottom, we show cases with one individual interaction, two simultaneous interactions, and the three interactions active at the same time.

Once the three pairwise interactions have been easily implemented in SymBoltz, we now consider three cases: a) only one active interaction at a time, b) two active interactions together by pairs and c) all interactions together. For that, we only need to tell the code the values of the coupling parameters $\alpha\mathtt{bc}$, $\alpha\mathtt{b}\Lambda$ and $\alpha\mathtt{c}\Lambda$ in the code. In Figure 12 we display the background evolution for the three interacting components considered here (dark matter, baryons and dark energy) and the Hubble rate for each of the three cases. In Figures 13 an 14 we display the matter power spectrum and the cosmic microwave background, respectively.

This model is proof of concept of the simplicity of implementing complicated models in SymBoltz. Here we have no corresponding implementation of CLASS to compare with, where it would take us more effort and care to implement this model.

VI. CONCLUSION

The main goal of this work is to show how the Einstein–Boltzmann code SymBoltz makes it particularly easy to implement interactions between baryons, dark matter and dark energy in a general way. Starting from SymBoltz' compact symbolic definition of the full standard ΛCDM model, we created a simple `interacting_model`$(Q_{ij}, \delta Q_{ij}, f_{ij})$ wrapper function around it that allows specifying arbitrary expressions for the energy/momentum flow between any

pair(s) of fluids (i, j). This makes it trivial to prototype arbitrary interactions between any fluids, and in a way where multiple interactions is no more complex than only one. Our full model definition is available in the code documentation[11], and is easy to modify.

Of course, the interaction should be physically well-defined. Here we showed how to derive them from a covariant model and in a way that conserves total energy-momentum. Once this is done, the implementation in SymBoltz is straightforward and only involves typing the symbolic expressions into the code.

The symbolic-numeric interface and approximation-free nature of SymBoltz makes it possible to declare all variables, parameters and equations of the cosmological model in a convenient, uniform and compact way. Support for Unicode characters makes the code equations readable and correspond one-to-one with the mathematical notation. We can specify all equations in a single place *as if* they are part of one system, but SymBoltz internally splits them into sequential background/perturbation stages with a mixture of forwards-backwards integration to solve them in the optimal way. The Rodas5P implicit solver coupled with efficient generated numerical code for the ODE and its sparse and analytical Jacobian is completely invisible to users, but ensures that the stiff equations are stable and fast to solve without introducing any approximation schemes.

Concretely, here we first validated a few models involving a single interaction between two species by comparing them to corresponding modified versions of CLASS from earlier works, finding 0.1%-1% difference for matter and CMB power spectra at most scales. We then constructed a proof-of-concept model with interactions between all three species at the same time, which was completely straightforward with our general interacting model constructor but would take more effort and care to implement in CLASS due to approximation schemes and more verbose boilerplate code for every operation in the code.

Giving the flexibility of SymBoltz, the current multi-interacting version of the code can be extended in two possible ways. First, we can consider an interaction that not only modifies the energy-momentum conservation rule but also the thermodynamical evolution of the interacting species. Having an effective modification on the temperature evolution equations of a interacting species can be added via a new variable, say Q_T, to the triad $(Q, \delta Q, f)$ for each interaction. With the goal of having, again, a very general and univocally defined approach, one could consider a framework in which SymBoltz, given the usual interaction triad $(\bar{Q}, \delta Q, f)$ together with the additional micro-physical information required to specify the exchange of internal energy, automatically incorporates the corresponding modifications into the temperature evolution equations through its symbolic-numeric framework. Second, we can easily incorporate new species by adding new variables and equations in our model definition, and adding its contribution to the total energy-momentum variables. However, our objective here is to test the ease of modifying the code, and we leave these generalization for future work.

Consequently, SymBoltz provides a direct route from any arbitrarily chosen covariantly specified interaction to full, approximation-free cosmological predictions. This makes the exploration of multi-component interacting cosmologies of any kind straightforward to code with systematic and quantitatively reliable calculations, with precision comparable to that of established state-of-the-art Boltzmann solvers.

[1] H. Sletmoen, Astron. Astrophys. **707**, A128 (2026), arXiv:2509.24740 [astro-ph.CO].
[2] J. Lesgourgues, arXiv e-prints , arXiv:1104.2932 (2011), arXiv:1104.2932 [astro-ph.IM].
[3] D. Blas, J. Lesgourgues, and T. Tram, **2011**, 034 (2011), arXiv:1104.2933 [astro-ph.CO].
[4] A. Lewis, JCAP **10**, 013 (2026), arXiv:2607.14854 [astro-ph.CO].
[5] J. Lesgourgues and T. Tram, **2011**, 032 (2011), arXiv:1104.2935 [astro-ph.CO].
[6] G. Steinebach, BIT Numerical Mathematics **63**, 27 (2023).
[7] C.-P. Ma and E. Bertschinger, Astrophys. J. **455**, 7 (1995), arXiv:astro-ph/9506072.
[8] M. B. Gavela, L. Lopez Honorez, O. Mena, and S. Rigolin, JCAP **11**, 044 (2010), arXiv:1005.0295 [astro-ph.CO].
[9] J. Valiviita, E. Majerotto, and R. Maartens, JCAP **07**, 020 (2008), arXiv:0804.0232 [astro-ph].
[10] R. von Marttens, L. Casarini, D. F. Mota, and W. Zimdahl, Phys. Dark Univ. **23**, 100248 (2019), arXiv:1807.11380 [astro-ph.CO].
[11] B. Wang, E. Abdalla, F. Atrio-Barandela, and D. Pavon, Rept. Prog. Phys. **79**, 096901 (2016), arXiv:1603.08299 [astro-ph.CO].
[12] B. Wang, E. Abdalla, F. Atrio-Barandela, and D. Pavón, Rept. Prog. Phys. **87**, 036901 (2024), arXiv:2402.00819 [astro-ph.CO].
[13] M. Chevallier and D. Polarski, International Journal of Modern Physics D **10**, 213 (2001), arXiv:gr-qc/0009008 [gr-qc].
[14] E. V. Linder, Phys. Rev. Lett. **90**, 091301 (2003), arXiv:astro-ph/0208512 [astro-ph].

[11] https://hersle.github.io/SymBoltz.jl/stable/interacting/

[15] F. Simpson, Phys. Rev. D **82**, 083505 (2010), arXiv:1007.1034 [astro-ph.CO].
[16] A. Pourtsidou, C. Skordis, and E. J. Copeland, Phys. Rev. D **88**, 083505 (2013), arXiv:1307.0458 [astro-ph.CO].
[17] C. G. Boehmer, N. Tamanini, and M. Wright, Phys. Rev. D **91**, 123003 (2015), arXiv:1502.04030 [gr-qc].
[18] A. Pourtsidou and T. Tram, Phys. Rev. D **94**, 043518 (2016), arXiv:1604.04222 [astro-ph.CO].
[19] M. S. Linton, A. Pourtsidou, R. Crittenden, and R. Maartens, JCAP **04**, 043 (2018), arXiv:1711.05196 [astro-ph.CO].
[20] F. N. Chamings, A. Avgoustidis, E. J. Copeland, A. M. Green, and A. Pourtsidou, Phys. Rev. D **101**, 043531 (2020), arXiv:1912.09858 [astro-ph.CO].
[21] L. Amendola and S. Tsujikawa, JCAP **06**, 020 (2020), arXiv:2003.02686 [gr-qc].
[22] J. Beltrán Jiménez, D. Bettoni, D. Figueruelo, F. A. Teppa Pannia, and S. Tsujikawa, JCAP **03**, 085 (2021), arXiv:2012.12204 [astro-ph.CO].
[23] D. Figueruelo *et al.*, JCAP **07**, 022 (2021), arXiv:2103.01571 [astro-ph.CO].
[24] M. S. Linton, R. Crittenden, and A. Pourtsidou, JCAP **08**, 075 (2022), arXiv:2107.03235 [astro-ph.CO].
[25] J. Beltrán Jiménez, D. Bettoni, D. Figueruelo, F. A. Teppa Pannia, and S. Tsujikawa, Phys. Rev. D **104**, 103503 (2021), arXiv:2106.11222 [astro-ph.CO].
[26] X. Liu, S. Tsujikawa, and K. Ichiki, Phys. Rev. D **109**, 043533 (2024), arXiv:2309.13946 [astro-ph.CO].
[27] N. Cruickshank, R. Crittenden, K. Koyama, and M. Bruni, (2025), arXiv:2512.11639 [astro-ph.CO].
[28] K. Aoki, J. Beltrán Jiménez, M. C. Pookkillath, and S. Tsujikawa, Phys. Rev. D **113**, 044053 (2026), arXiv:2504.17293 [astro-ph.CO].
[29] E. Jensko, E. M. Teixeira, and V. Poulin, (2026), arXiv:2603.10622 [astro-ph.CO].
[30] J. Beltrán Jiménez, K. Ichiki, X. Liu, F. A. Teppa Pannia, and S. Tsujikawa, (2026), arXiv:2603.15805 [astro-ph.CO].
[31] M. Asghari, J. Beltrán Jiménez, S. Khosravi, and D. F. Mota, JCAP **04**, 042 (2019), arXiv:1902.05532 [astro-ph.CO].
[32] J. Beltrán Jiménez, D. Bettoni, D. Figueruelo, and F. A. Teppa Pannia, JCAP **08**, 020 (2020), arXiv:2004.14661 [astro-ph.CO].
[33] D. Figueruelo and F. A. Teppa Pannia, (2026), arXiv:2607.22415 [astro-ph.CO].
[34] F. Ferlito, S. Vagnozzi, D. F. Mota, and M. Baldi, Mon. Not. Roy. Astron. Soc. **512**, 1885 (2022), arXiv:2201.04528 [astro-ph.CO].
[35] M. van der Westhuizen, A. Abebe, and E. Di Valentino, Phys. Dark Univ. **50**, 102119 (2025), arXiv:2509.04495 [gr-qc].
[36] J.-H. He, B. Wang, and E. Abdalla, Phys. Rev. D **83**, 063515 (2011), arXiv:1012.3904 [astro-ph.CO].
[37] A. A. Costa, X.-D. Xu, B. Wang, E. G. M. Ferreira, and E. Abdalla, Phys. Rev. D **89**, 103531 (2014).
[38] M. A. van der Westhuizen and A. Abebe, JCAP **01**, 048 (2024), arXiv:2302.11949 [gr-qc].
[39] K. R. Mishra, S. K. J. Pacif, R. Kumar, and K. Bamba, Phys. Dark Univ. **40**, 101211 (2023), arXiv:2301.08743 [gr-qc].
[40] M. van der Westhuizen, A. Abebe, and E. Di Valentino, Phys. Dark Univ. **50**, 102120 (2025), arXiv:2509.04494 [gr-qc].
[41] Y. L. Bolotin, A. Kostenko, O. A. Lemets, and D. A. Yerokhin, Int. J. Mod. Phys. D **24**, 1530007 (2014), arXiv:1310.0085 [astro-ph.CO].
[42] M. Bouhmadi-López, J. Morais, and A. Zhuk, Phys. Dark Univ. **14**, 11 (2016), arXiv:1603.06983 [gr-qc].